Chapter # 09
Immunity

Student Learning Outcomes

After studying this chapter, students will be able to:

- [B-12-1-01] List the structural features of human skin that make it an impenetrable barrier ' against invasion by microbes. (1st line of defence).
- [B-12-1-02] Explain how oil and sweat glands within the epidermis inhibit the growth and also kill microorganisms. (1st line of defence).
- [B-12-1-03] Recognize the role of the acids of the digestive tract as killing bacteria present in food.
- [B-12-1-04] State the role of the ciliated epithelium of the nasal cavity and the mucous of the bronchi and bronchioles in trapping airborne microorganisms.
- [B-12-1-05] Describe the role of macrophages and neutrophils in killing bacteria.
- [B-12-1-06] Explain how Natural Killer (NK) cells kill cells infected by microbes and cancer cells.
- [B-12-1-07] State the way proteins of the complement system kill bacteria and that interferons inhibit viruses from infecting cells.
- [B-12-1-08] State the events of the inflammatory response as a generalized, nonspecific defence.
- [B-12-1-09] Outline the release of pyrogens by microbes and their effect on the hypothalamus to boost the body's temperature.
- [B-12-1-10] List the ways that fever affects microbes.
- [B-12-1-11] Define the specific immune system as providing specific defence and acting as the most powerful means of resisting infection.
- [B-12-1-12] Identify monocytes, T cells, and B-cells as components of the immune system.
- [B-12-1-13] State inborn and acquired immunity as the two basic types of immunity.
- [B-12-1-14] Differentiate between active and passive immunity as the two types of acquired immunity.
- [B-12-1-15] Describe the role of T-cells in cell-mediated immunity.
- [B-12-I-16] Describe the role of B-cells in antibody-mediated immunity.
- [B-12-1-17] Discuss the role of T -cells and B -cells in transplant rejections.
- [B-12-1-18] Evaluate the discovery of monoclonal antibodies and justify how this accomplishment revolutionized many aspects of biological research.
- [B-12-I-19] Identify the process of vaccination as a means to develop active acquired immunity.
- [B-12-1-20] Draw the structural model of an antibody molecule.
- [B-12-I-21] Explain the role of memory cells in long term immunity.
- [B-12-1-22] Define allergies and correlate the symptoms of allergies with the release of histamines.
- [B-12-1-23] Describe the autoimmune diseases with examples.

Every living organism is constantly exposed to a variety of harmful agents in its environment, such as bacteria, viruses, fungi, parasites, and toxins. To survive and remain healthy, the body must defend itself against these threats. This defense is carried out by a highly coordinated system known as the ***immune system***, and the protective response it provides is called **Immunity**.

Immunity ensures that the body can **recognize, neutralize, and eliminate** harmful substances without damaging its own healthy cells. A key feature of this system is its ability to distinguish between **self** (the body's own cells and molecules) and **non-self** (foreign invaders). This recognition is vital, as it prevents unwanted attacks on the body's own tissues.

The immune system employs different lines of defense. The **first line of defense** includes physical and chemical barriers such as skin, mucous membranes, and secretions that prevent pathogens from entering the body. If these barriers are breached, the **second line of defense** provides rapid but non-specific responses, such as inflammation and phagocytosis. Beyond these, the body also possesses a **third line of defense**, the adaptive immune system, which produces highly specific responses involving lymphocytes and antibodies. Importantly, adaptive immunity has **memory**, meaning it responds more quickly and strongly to repeated infections by the same pathogen.

Immunity can be classified as **innate** (present from birth, providing general defense) or **acquired** (developed after exposure to pathogens or through vaccination). Together, these mechanisms protect the body from disease, ensure recovery after infection, and contribute to long-term survival.

The study of immunity not only explains how the body defends itself but also underlies important applications such as vaccines, allergy treatments, and organ transplantation. Understanding this system is therefore crucial for both biology and medicine.

IMMUNITY

Living organisms are constantly exposed to disease-causing agents such as bacteria, viruses, fungi, parasites, and their toxins. To survive, the human body has evolved complex defence mechanisms that prevent or fight infections. The study of these protective processes is known as **immunology**, while the protective ability itself is called **immunity**.

Immunity

Immunity is defined as the body's ability to resist or eliminate damage caused by harmful microbes or toxic substances. It ensures that invading pathogens are destroyed before they can cause disease.

Immune Response

The **immune response** refers to the body's coordinated reaction to foreign substances (antigens). This may involve non-specific actions such as inflammation or phagocytosis, or highly specific responses like the production of antibodies by lymphocytes.

Lines of Defence in Humans

The human body protects itself through **three lines of defence:**

1. **First Line of Defence (Physical and Chemical Barriers)**
   - This includes the **skin** and **mucous membranes**, which act as physical barriers to block the entry of pathogens.
   - The skin is the most important structural defence, providing an almost impenetrable shield. Its outermost layer, the **epidermis**, is composed of tightly packed cells and a protective layer of dead, keratinized cells that resist microbial invasion.
   - *Additional* protective features of the skin include:
     - Sebum (oil secreted by sebaceous glands), which lowers skin pH and inhibits microbial growth.
     - **Sweat**, containing lysozyme and salts, which break down bacterial cell walls and create unfavorable conditions for microbes.
     - ***Normal skin flora***, beneficial microbes that outcompete harmful pathogens.
   - Together, these features make the skin a powerful barrier and the body's **first line of defence** against infection.
2. **Second Line of Defence (Non-Specific Internal Responses)**
   - Activated if pathogens bypass physical barriers.
   - Includes processes such as **phagocytosis, inflammation, fever**, and the activity of certain white blood cells.

3. Third Line of Defence (Specific/Adaptive Immunity)

- Provides highly targeted responses against specific pathogens.
- Involves **lymphocytes (B and T cells)** and the production of **antibodies.**
- Creates **immunological memory**, ensuring faster and stronger responses during repeated infections.

▶ Innate vs. Specific Immunity

- The **first and second lines of defence** together form the **innate immune system**, which provides immediate but non-specific protection.
- The **third line of defence** is the **specific (adaptive) immune system**, capable of precision targeting and long-term memory.

IMMUNE RESPONSE

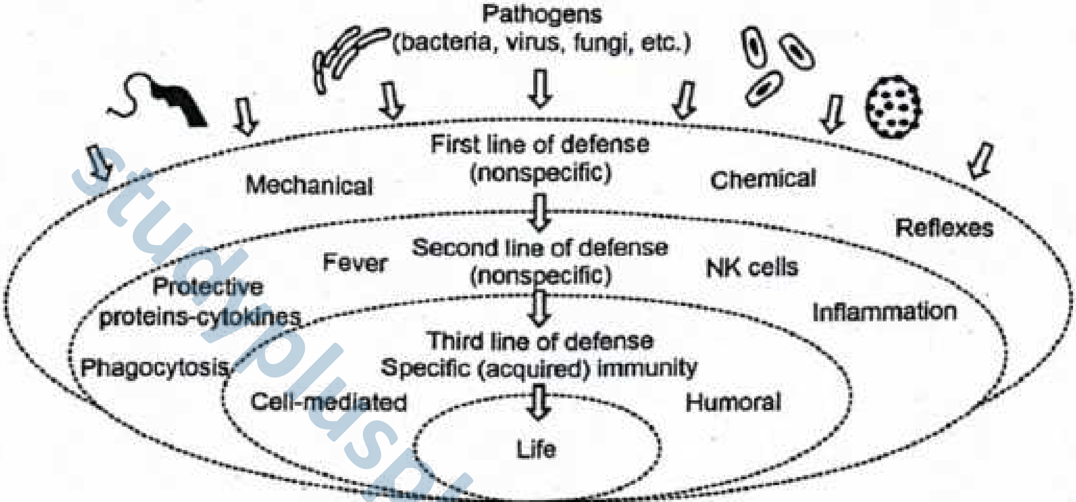

Figure: The five drug development phases

> **SLO [B-12-1-01]**
> **List the structural features of human skin that make it an impenetrable barrier ' against invasion by microbes. (1st line of defence).**

SKIN: AN IMPENETRABLE BARRIER AGAINST MICROBIAL INVASION

The **skin** serves as the body's largest organ and a critical component of the **first line of defence.** It acts as a **multilayered barrier** against microbial invasion through a combination of **physical, chemical, and immunological mechanisms.**

Structurally, the skin is composed of two primary layers:

1. Epidermis

The **epidermis** is the outermost layer, consisting of **tightly packed epithelial *cells*** that form a robust **physical barrier** against pathogen entry. Most epidermal cells produce **keratin,** a fibrous protein that provides *mechanical* strength and reduces **water loss,** further limiting microbial survival. The skin's outer surface is composed primarily of **dead, keratinized cells,** creating a **dry environment** that is inhospitable to most microorganisms, which require moisture and nutrients for survival.

The skin also maintains a **slightly acidic pH (~5.5)** due to secretions from sebaceous and sweat glands. This acidity inhibits the growth of many bacteria and fungi. Additionally, the process of **desquamation**—the continuous shedding of dead skin cells—helps remove surface microbes that may have adhered to the skin.

2. Dermis

Beneath the epidermis lies the **dermis,** a thicker, vascularized layer containing **hair follicles, nerve endings,** glands, and **blood vessels.** Within the dermis, **sebaceous glands** secrete **sebum,** an oily substance rich in fatty acids that lower skin pH and exhibit **antimicrobial activity.**

Sweat glands produce **sweat,** a watery secretion that plays a thermoregulatory role but also contributes to defence. Sweat contains **lysozyme,** an enzyme capable of degrading **bacterial cell walls,** and has a **high salt concentration,** which can **dehydrate bacteria** and inhibit their growth.
The combined action of **sebum and sweat** forms a protective chemical layer over the skin. These secretions contain **natural antimicrobial agents,** such as **lactic acid,** that deter colonization by harmful microbes.

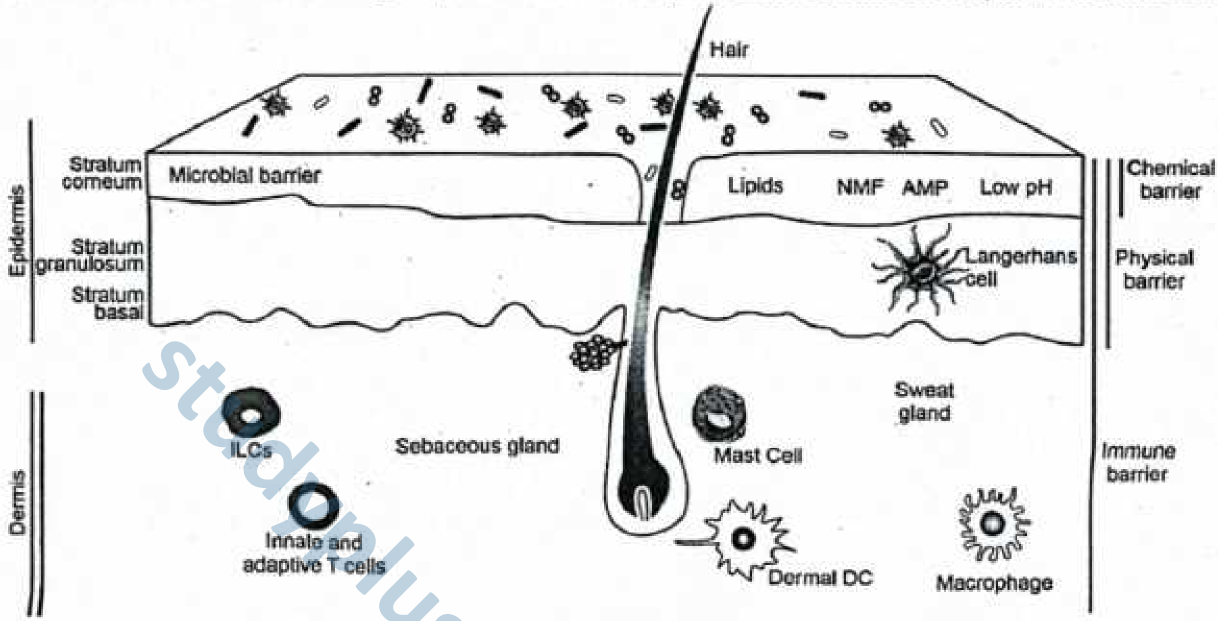

Figure: Skin as an impenetrable barrier against microbial invasion

> **SLO [B-12-1-02]**
> **Explain how oil and sweat glands within the epidermis inhibit the growth and also kill microorganisms. (1st line of defence).**

ROLE OF OIL AND SWEAT GLANDS IN DEFENCE AGAINST MICROORGANISMS

The **epidermis** of human skin is not only a physical barrier but also an active chemical defence system. Two specialized exocrine glands—**sebaceous (oil) glands** and **sweat glands**—secrete substances that inhibit microbial growth and, in many cases, directly kill pathogens.

Role of Oil and Sweat Glands in the First Line of Defence

The skin is not only a physical shield against microbial invasion but also a chemical barrier. Two specialized glands within the epidermis—the **sebaceous (oil) glands** and the **sweat glands**—produce secretions that inhibit or kill microorganisms, making the skin a powerful part of the body's first line of defence.

1. **Sebaceous (Oil) Glands**

   The sebaceous glands secrete an oily substance called **sebum** onto the skin surface. This secretion has multiple protective roles:

   - **Antimicrobial Action:** Sebum contains **fatty acids** that lower the skin's pH to about **4–6**, *creating an* acidic environment that is unfavorable for the survival of many bacteria and fungi. The oily layer also prevents microbes from adhering to and colonizing the skin surface. In addition, certain components of sebum have **direct bactericidal properties,** capable of destroying harmful microorganisms.
   - ***Moisturizing Effect:*** Sebum keeps the skin **soft and supple,** reducing the risk of cracks and abrasions that could otherwise serve as entry points for pathogens.

2. **Sweat Glands**

   The sweat glands release **sweat (perspiration)** through pores on the skin. Sweat strengthens the skin's defence in the following ways:

- **Antimicrobial Action:** Sweat contains **lysozyme**, an enzyme that breaks down the peptidoglycan of bacterial cell walls, leading to bacterial lysis. Additionally, the **high salt concentration** in sweat creates **osmotic stress**, dehydrating and killing many microbes.
- **Mechanical Cleansing:** Sweat also helps to **wash away pathogens** from the skin surface, reducing the likelihood of colonization.

Together, **oil and sweat glands** create an unfavorable environment for invading microbes by:

- Lowering skin pH (acidic conditions).
- Producing antimicrobial substances like fatty acids and lysozyme.
- Increasing salt concentration, leading to dehydration of microbes.
- Reducing microbial colonization on the skin surface.

Thus, the combined action of oil and sweat glands strengthens the **first line of defence**, ensuring that the skin acts not only as a **physical barrier** but also as a **chemical shield** against infections.

SLO [B-12-1-03]
Recognize the role of the acids of the digestive tract as killing bacteria present in food.

ROLE OF DIGESTIVE TRACT ACIDS IN KILLING BACTERIA

The digestive tract is constantly exposed to foreign substances through the intake of food and water. Along with nutrients, these ingested materials often carry **microorganisms**, including bacteria, fungi, and sometimes viruses. To protect the body from infection, the digestive system is equipped with strong **chemical barriers**, the most important of which are **gastric acids**.

1. Hydrochloric Acid in the Stomach

One of the most important chemical barriers of the digestive tract is **hydrochloric acid (HCl),** secreted by the **parietal cells** of gastric glands in the stomach lining. Its presence ensures that food entering the stomach is not only digested but also disinfected from harmful microorganisms.

- **Low pH Environment:** Hydrochloric acid lowers the stomach pH to around **1.5–3.5**, creating a highly acidic environment. Such conditions are lethal for most bacteria, fungi, and other microbes *ingested* with food and water.
- **Denaturation of Proteins and Microbial Structures:** The strong acidity denatures proteins, unfolding them and disrupting their structure. This effect also damages microbial **cell walls and plasma membranes**, leading to the inactivation or death of pathogens.
- **Activation of Digestive Enzymes:** Hydrochloric acid converts **pepsinogen** (inactive form) into **pepsin**, a powerful enzyme that digests proteins. This enzymatic breakdown further disrupts microbial proteins, weakening their ability to survive in the stomach.

Together, these functions make gastric hydrochloric acid a **critical defence mechanism** of the digestive system, preventing most harmful microbes from reaching the intestines and causing infection.

2. Protective Role against Foodborne Pathogens

Food and water often contain harmful microorganisms that can cause serious gastrointestinal diseases if they are not neutralized. The **acidic environment of the stomach** plays a vital role in preventing these infections by killing or inactivating most pathogens before they reach the intestines.

- **Destruction of Foodborne Bacteria:** Many common foodborne microbes, including ***Escherichia coli***, ***Salmonella***, and ***Vibrio cholerae***, cannot survive prolonged exposure to gastric acid. The extremely low pH disrupts their cellular structures and halts their metabolic processes, effectively eliminating them.
- **Prevention of Intestinal Infections:** By destroying harmful organisms at the stomach level, hydrochloric acid prevents them from entering the intestines, where conditions are more favorable for microbial growth. Without this protective mechanism, these pathogens could multiply rapidly, leading to food poisoning, diarrhea, or more severe infections.
- **First Chemical Barrier:** In this way, gastric acid serves as an essential **first chemical barrier** in the digestive tract. It ensures that most ingested microbes are neutralized early, reducing the risk of disease and safeguarding the body's internal environment.

3. Variations in Acid Protection

The protective role of gastric acid can vary among individuals depending on the amount of acid secreted in the stomach. This directly influences the body's ability to defend itself against foodborne pathogens.

- **High Acidity (Normal Secretion):** In healthy individuals with normal gastric acid secretion, the stomach maintains a highly acidic environment (pH 1.5–3.5). This provides strong protection by killing most ingested bacteria and preventing infections.
- **Reduced Acidity (Hypochlorhydria or Achlorhydria):** In some individuals, the production of stomach acid is significantly reduced.
  - **Hypochlorhydria** refers to abnormally low levels of gastric acid.
  - **Achlorhydria** refers to a complete absence of gastric acid secretion. These conditions may arise due to aging, long-term use of antacids or proton pump inhibitors, or certain medical disorders.
- **Consequences of Low Acidity:** When acidity is reduced, many pathogens that would normally be destroyed can survive the stomach environment and reach the intestines. This increases susceptibility to gastrointestinal infections, food poisoning, and other digestive disorders.

4. Additional Acidic Secretions in the Digestive Tract

While hydrochloric acid in the stomach is the primary antimicrobial factor, other parts of the digestive system also contribute:

- **Fatty acids and bile salts** in the small intestine interfere with bacterial membranes.
- The **acidic environment of the stomach** works synergistically with **digestive enzymes and intestinal secretions** to reduce microbial survival.

The acids of the digestive tract, especially **hydrochloric acid secreted in the stomach**, serve as a powerful barrier against infection by:

- Creating an acidic environment hostile to microbes.
- Denaturing microbial proteins and damaging cell structures.
- Killing or inactivating many bacteria present in food and water.

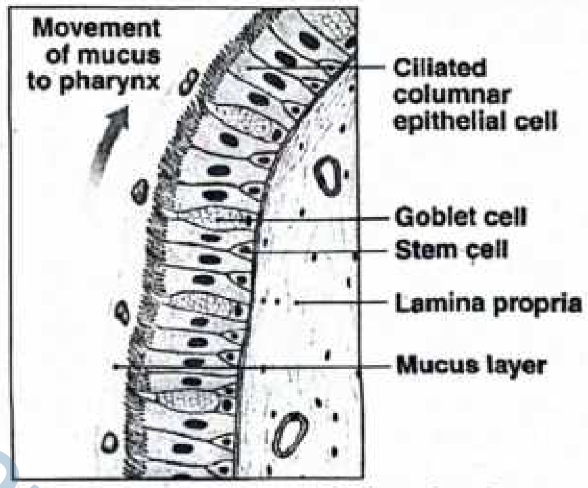

Figure: Respiratory epithelium of trachea

Thus, gastric acids form an essential part of the **body's first line of defence**, ensuring that harmful microbes rarely reach the intestines to cause disease.

> **SLO [B-12-1-04]**
> **State the role of the ciliated epithelium of the nasal cavity and the mucous of the bronchi and bronchioles in trapping airborne microorganisms.**

ROLE OF THE RESPIRATORY TRACT IN TRAPPING AIRBORNE MICROORGANISMS

The respiratory system is constantly exposed to the external environment, making it vulnerable to airborne dust, allergens, and microorganisms. To protect against these threats, specialized structural features such as **ciliated epithelium** and **mucus secretions** form part of the body's **first line of defence**.

1. Ciliated Epithelium of the Nasal Cavity

The **nasal cavity** is the first part of the respiratory tract that comes into contact with inhaled air. Its inner surface is lined with a **ciliated epithelium**, a specialized tissue made up of epithelial cells bearing numerous microscopic, hair-like projections called **cilia.**

- **Structure of Cilia:** These cilia are fine, flexible structures that extend outward from the cell surface. They are arranged in dense rows and are capable of beating in a **rhythmic and coordinated fashion.**

- **Role in Defence:** As air enters the nasal passages, it often carries dust, pollen, and airborne microorganisms such as bacteria, fungal spores, and viruses. These particles become trapped in the thin layer of **mucus** that coats the nasal lining.
- **Mechanism of Action:** The rhythmic beating of cilia acts like a conveyor belt, pushing the mucus, along with the trapped microbes and dust particles, slowly backward towards the **pharynx**.
- **Final Clearance:** Once in the pharynx, the mucus is either:
  - **Swallowed**, so that the microbes are destroyed by the strong **hydrochloric acid** in the stomach, or
  - **Expelled** through defensive reflexes such as **sneezing** or **coughing**.

Thus, the ciliated epithelium of the nasal cavity provides a **mechanical barrier** that prevents harmful microorganisms from reaching the lower respiratory tract and causing infections.

2. Mucus of the Bronchi and Bronchioles

The **bronchi and bronchioles** form the conducting airways that carry air deeper into the lungs. These passageways are lined with epithelial cells, including **goblet cells** and **mucous glands**, which secrete a thick and sticky fluid known as **mucus**.

- **Composition and Properties:** The mucus is rich in glycoproteins, which give it a viscous and adhesive nature. This sticky layer acts as a **biological trap** for inhaled particles such as dust, pollen, allergens, and, most importantly, airborne microorganisms like bacteria, viruses, and fungal spores.
- **Protective Coating:** By coating the internal lining of the bronchi and bronchioles, mucus prevents pathogens from coming into direct contact with the delicate epithelial tissues, thereby acting as the **first barrier within the lower respiratory tract**.
- **Mucociliary Escalator:** Once pathogens and debris are trapped, the **cilia** lining the bronchi and bronchioles beat in a coordinated upward motion. This continuous movement, often called the **mucociliary escalator**, carries the mucus layer together with its trapped particles toward the **pharynx**.
- **Final Removal:** In the throat, the mucus may either **be swallowed**, allowing stomach acid to destroy the pathogens, or **expelled** by coughing, sneezing, or spitting.
- **Functional Significance:** This protective mechanism is crucial in ensuring that harmful microorganisms and irritants do not reach the **alveoli**, the delicate air sacs of the lungs where gas exchange occurs. Protection of the alveoli is vital because even minor infections at this site can severely impair breathing efficiency.

Thus, the mucus of the bronchi and bronchioles, in coordination with ciliary action, provides an **active defence system** that maintains the sterility of the lower respiratory tract and safeguards efficient respiration.

Together, the **ciliated epithelium** and **mucus secretions** provide an effective mechanical barrier against airborne microorganisms. They work to trap, immobilize, and expel pathogens before these can establish infections in the respiratory tract.

SLO [B-12-1-05]
Describe the role of macrophages and neutrophils in killing bacteria.

When pathogens successfully cross the **first line of defence** (skin, mucous membranes, and secretions), the body activates the **second line of defence**. This line consists of **non-specific internal responses** that aim to eliminate invaders regardless of their identity. Among these defences, a critical role is played by **phagocytic and killing cells of the blood**, which actively engulf and destroy invading microbes. The two most important phagocytes are **macrophages** and **neutrophils.**

Macrophages and Dendritic Cells

Both **macrophages** and **dendritic cells** are specialized phagocytic cells that arise from **monocytes**, a type of white blood cell produced in the **bone marrow**. After circulating in the blood for 1–3 days, monocytes migrate into tissues, where they differentiate into these long-lived defence cells. They play a central role in the **second line of defence** by engulfing microbes and linking the **innate immune system** with *the* **adaptive immune system.**

Functions of Macrophages

1. ***Distribution and Presence***

   Macrophages are widely distributed throughout the body, particularly in tissues such as the **lungs, liver, spleen, kidneys, and lymph nodes.** They are **long-lived cells** that act as permanent *sentinels of the immune* system. By patrolling interstitial spaces, they continuously monitor for invading pathogens, damaged cells, and cellular debris. This widespread distribution ensures that infections can be detected and dealt with rapidly at multiple sites in the body.

2. **Phagocytosis**

The primary role of macrophages is **phagocytosis**, the process of engulfing and destroying harmful microbes. They extend **cytoplasmic projections (pseudopodia)** around a bacterium or particle, enclosing it in a vesicle called a **phagosome**. The phagosome then fuses with lysosomes to form a **phagolysosome**. Inside this structure, destructive enzymes and highly reactive oxygen species break down the microbe, effectively neutralizing the threat.

3. **Antigen Presentation**

In addition to killing bacteria, macrophages serve as **antigen-presenting cells (APCs)**. After digestion, they process microbial fragments and display them on their cell surface bound to special proteins (MHC molecules). These presented antigens are recognized by **T lymphocytes**, which become activated and launch a **specific adaptive immune response**. Through this function, macrophages act as a bridge between **innate immunity** and **adaptive immunity**.

4. **Secretion of Signalling Molecules**

Macrophages also play a vital role in coordinating immune responses through the secretion of **signalling molecules**:

- **Cytokines** are released to recruit and activate more immune cells at the site of infection.
- **Interleukin-1 (IL-1)** is secreted to act on the hypothalamus, raising body temperature and producing fever, which slows bacterial growth.
- **Growth factors** secreted by macrophages stimulate **tissue repair and wound healing** once the infection is under control.

Functions of Dendritic Cells

1. **Location**

Dendritic cells are found mainly in **epithelial tissues** that are directly exposed to the external environment, making them one of the first immune cells to encounter invading pathogens. They are present in the **skin** (where they are known as *Langerhans cells*), the **respiratory tract mucosa**, and the **intestinal lining**. Their strategic positioning allows them to detect microbes at common entry points into the body.

2. **Phagocytosis and Antigen-Presenting Role**

Like macrophages, dendritic cells are capable of **phagocytosis**, engulfing bacteria, viruses, and other antigens. However, they are regarded as **professional antigen-presenting cells (APCs)** because they are exceptionally efficient in **processing and displaying microbial antigens** on their surface using MHC molecules. This ability makes them one of the most critical immune cells for communication between the innate and adaptive immune systems.

3. **Activation of Adaptive Immunity**

A distinctive feature of dendritic cells is their ability to **migrate**. After capturing antigens in peripheral tissues, they travel to nearby **lymph nodes**, where they present the processed antigens to **naïve T lymphocytes**. This interaction is crucial for initiating adaptive immune responses. By activating T cells, dendritic cells not only trigger immediate defense mechanisms but also help establish **immunological memory**, ensuring long-term protection against future infections by the same pathogen.

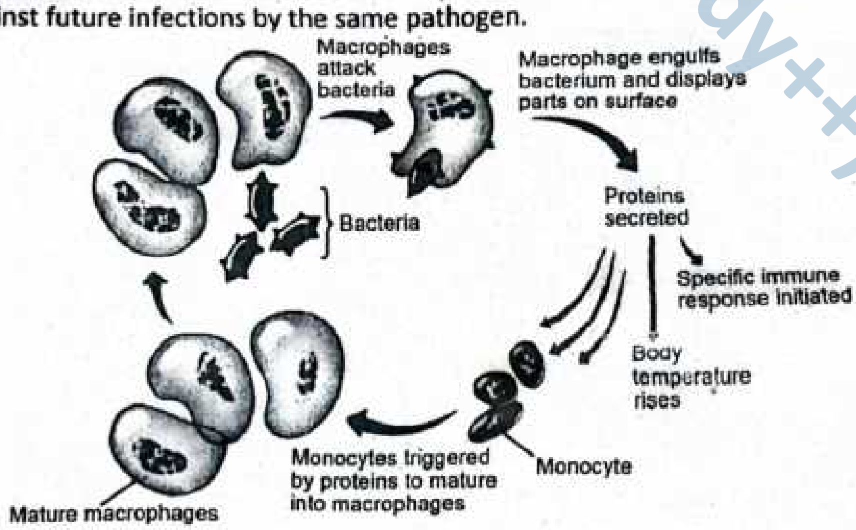

Figure: Response of macrophages to foreign particles

Neutrophils

Neutrophils are **granulocytic white blood cells (WBCs)** and represent the most abundant type of phagocytes circulating in the blood. They are **short-lived but highly motile**, making them one of the first immune cells to reach the site of infection. Their rapid response allows them to serve as the **frontline defenders** in the second line of defence.

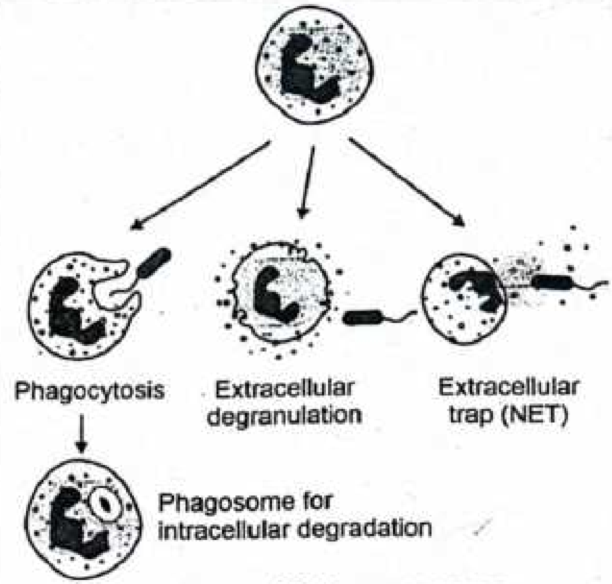

Figure: Neutrophils immune response

1. **Abundance and Rapid Response**

   Neutrophils are the **most numerous WBCs** in human circulation. They are the **first wave of phagocytes** recruited to an infection site. Unlike macrophages, which are long-lived, neutrophils survive only a short time but act as **highly efficient killers** during acute infections.

2. **Amoeboid Movement and Diapedesis**

   Neutrophils display **amoeboid movement**, which enables them to actively move through tissues. They can undergo **diapedesis**, a process where they squeeze through the walls of capillaries to reach infected or inflamed tissues quickly.

3. **Phagocytosis**

   Once at the infection site, neutrophils engulf pathogens into **phagosomes**, which then fuse with lysosomes. This fusion releases **digestive enzymes, antimicrobial peptides, and reactive oxygen species (ROS)** that effectively destroy the microbes.

4. **Degranulation**

   Neutrophils can also undergo **degranulation**, releasing stored antimicrobial substances directly into the extracellular space. These toxic compounds attack and neutralize nearby microbes without the need for engulfment.

5. **Neutrophil Extracellular Traps (NETs)**

   Apart from phagocytosis, neutrophils can release **Neutrophil Extracellular Traps (NETs)**, which are web-like structures made of DNA fibers coated with antimicrobial proteins. NETs immobilize and kill pathogens outside the cell, preventing their spread.

6. **Role in Inflammation and Pus Formation**

   During infection, neutrophils contribute to the **inflammatory response** by releasing enzymes and signaling molecules. Their accumulation at infection sites leads to the formation of **pus**, which consists largely of dead neutrophils, bacterial fragments, and tissue debris.

Together, macrophages and neutrophils form the core cellular defence of the second line of defence. Macrophages provide sustained phagocytosis and immune signalling, while neutrophils act as rapid, short-lived bacterial killers. Their combined activity ensures that bacterial invaders are destroyed quickly, preventing the spread of infection and buying time for the activation of the third line of defence (adaptive immunity).

DO YOU KNOW?

Do you know that Dendritic cells, macrophages, and B cells can function as antigen-presenting cells (APCs)?

Antigen-presenting cells (APCs): These are immune cells that process antigens and present them on their surface bound to **MHC (Major Histocompatibility Complex) molecules** to activate T-cells.

- **Dendritic cells** – the most potent APCs, specialized in capturing antigens and migrating to lymph nodes to activate naïve T-cells.
- **Macrophages** – phagocytose pathogens and present their antigens to helper T-cells, especially during ongoing infections.
- **B cells** – recognize specific antigens via their surface immunoglobulins, internalize them, and present processed peptides to helper T-cells.

All three—dendritic cells, macrophages, and B cells—are professional APCs because they not only present antigens but also provide the **co-stimulatory signals** required for T-cell activation.

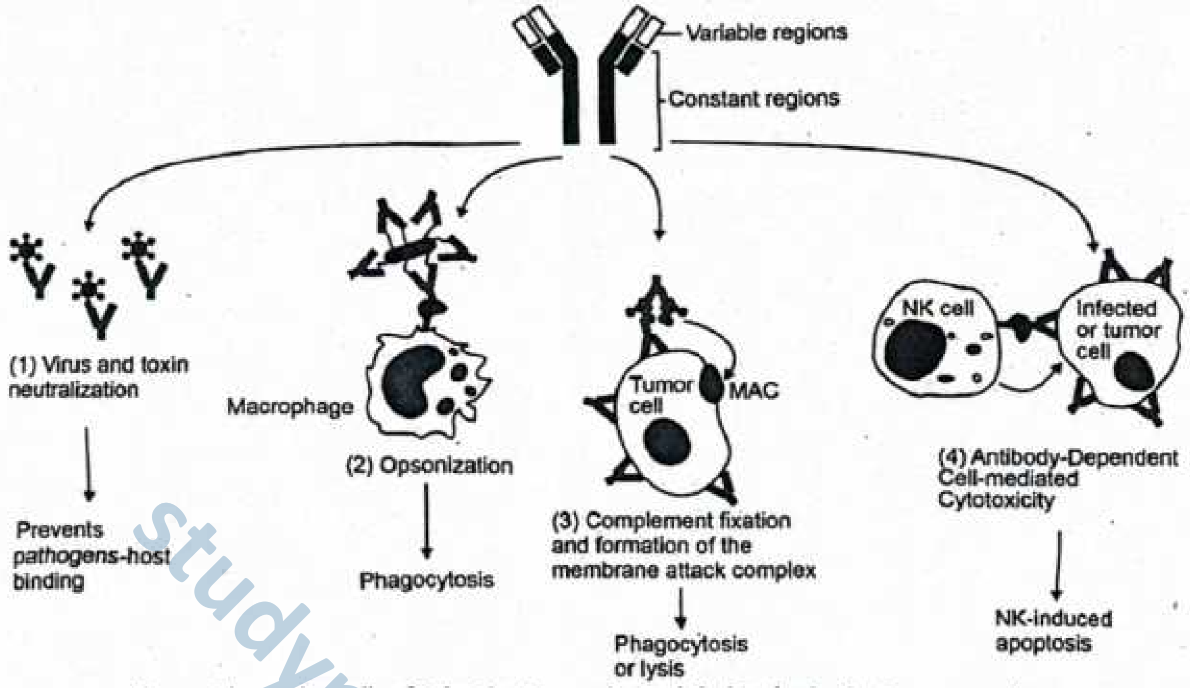

Figure: The major cells of Adaptive Immunity and their roles in the immune system

SLO [B-12-1-06]
Explain how Natural Killer (NK) cells kill cells infected by microbes and cancer cells.

NATURAL KILLER (NK) CELLS

Natural Killer (NK) cells are **large granular lymphocytes** that form an essential part of the **second line of defence** in the innate immune system. Unlike phagocytes, which engulf microbes, NK cells directly target and kill **virus-infected cells** and **cancer (tumour) cells** without prior sensitization.

Mechanism of Action

1. **Recognition of Abnormal Cells:**

   Natural Killer (NK) cells constantly patrol the body, scanning tissues for infected or cancerous host cells. Unlike cytotoxic T lymphocytes, which require antigen presentation through MHC molecules, NK cells recognize abnormalities more directly. Healthy cells normally display sufficient levels of MHC class I proteins on their surface, acting as "self-markers." However, many virus-infected and tumour cells reduce or lose these MHC class I molecules or present unusual stress-induced ligands. NK cells are able to detect these changes, allowing them to distinguish abnormal cells from healthy ones and target them for destruction.

2. **Release of Cytotoxic Molecules**

   Once an abnormal cell is recognized, NK cells form a specialized contact point known as the immunological synapse and release toxic granules stored in their cytoplasm. These granules contain perforins and granzymes, which work together to induce cell death. Perforins insert themselves into the plasma membrane of the target cell, where they assemble into pores that compromise membrane integrity. Through these pores, granzymes enter the cell's cytoplasm and initiate a cascade of molecular events that trigger apoptosis, or programmed cell death. This controlled mechanism ensures that the infected or tumour cell is dismantled without causing extensive damage to surrounding tissues.

3. **Removal of Dead Cells**

   Following apoptosis, the dying cell breaks apart into smaller membrane-bound fragments called apoptotic bodies. These fragments are then efficiently removed by macrophages and other phagocytic cells. This process not only prevents the release of harmful intracellular substances that could damage nearby tissues but also ensures that pathogens do not spread further within the body.

NK cells act **rapidly and non-specifically**, providing immediate protection while the adaptive immune system is still being activated. This makes them vital for early defence against **viral infections** and **tumour formation.**

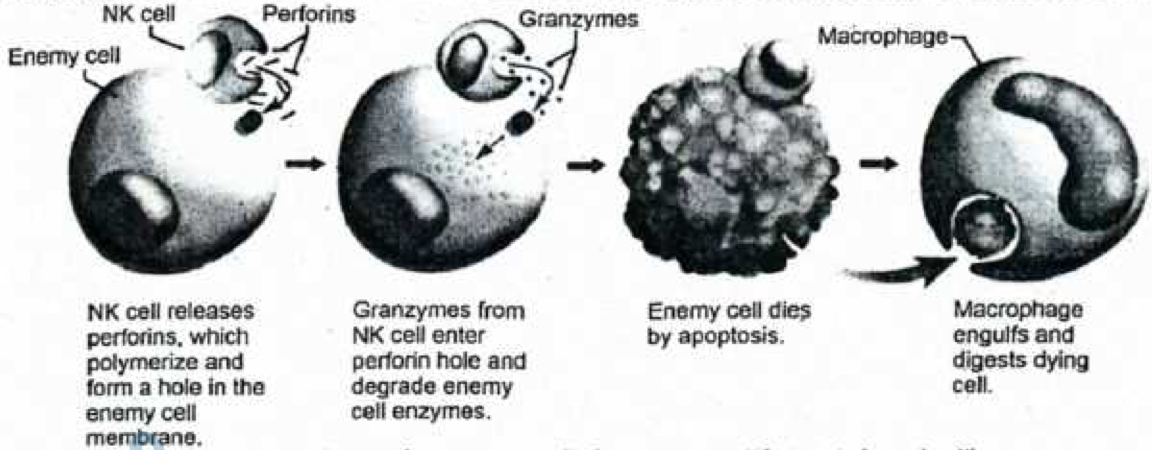

Figure: NK cells attack on enemy cells (Cancerous cell/ Virus infected cell)

SLO [B-12-1-07]
State the way proteins of the complement system kill bacteria and that interferons inhibit viruses from infecting cells.

PROTECTIVE PROTEINS OF COMPLEMENT SYSTEM

Complement System

The **complement system** is a cascade of over 50 small proteins circulating in the blood in an inactive form. These proteins are primarily produced by the liver and are activated upon detection of pathogens. Once initiated, the cascade amplifies the response in a chain-reaction manner.

Key outcomes of complement activation include:

- **Opsonization:** Coating of microbes to enhance phagocytosis.
- **Chemotaxis**: Recruitment of phagocytic cells to the infection site.
- **Formation of Membrane Attack Complexes (MACs):** A ring of proteins that inserts into microbial membranes, forming pores that disrupt osmotic balance, leading to cell lysis.
- **Promotion of inflammation**: Complement fragments stimulate vascular permeability and leukocyte activation.

The complement system thus bridges innate and adaptive immunity and enhances the effectiveness of both.

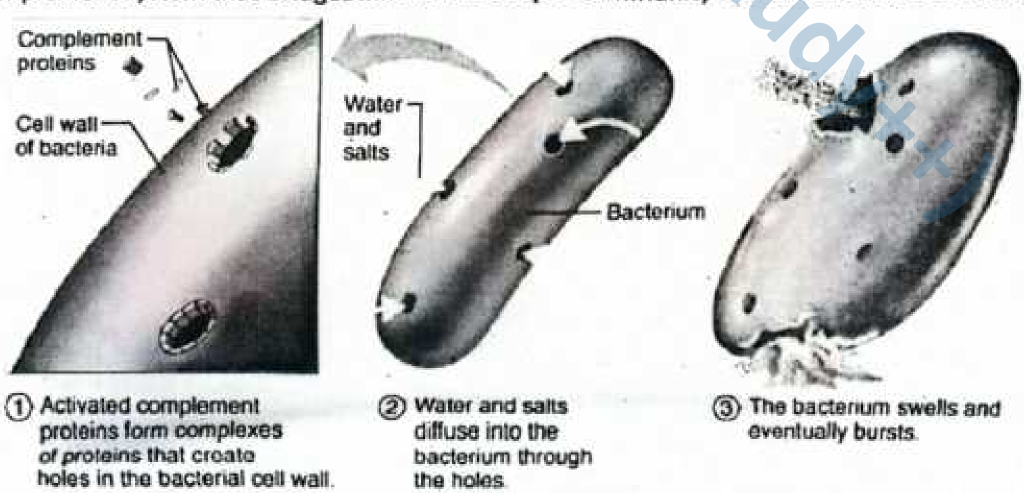

Figure: Action of the complement system against a bacterium

Interferons

Interferons (IFNs) are cytokines released primarily by virus-infected cells. They play a crucial role in limiting viral replication and alerting neighbouring cells to the viral threat.

Major actions of interferons include:

- **Antiviral state induction:** Stimulate neighbouring cells to produce antiviral proteins that degrade viral RNA and inhibit viral protein synthesis.
- **Immune cell activation:** Stimulate natural killer cells and macrophages.
- **Apoptosis induction:** Promote programmed death of infected cells.
- **Inhibition of viral release:** Restrict budding of new viruses from host cells.

Different types of interferons (e.g., IFN-α, IFN-β, IFN-γ) perform varied functions, including roles in bacterial and parasitic infections, immune regulation, and modulation of cell growth and differentiation.

DO YOU KNOW?

A cytokine is a chemical messenger that regulates cell differentiation, production, and gene expression to affect immune responses. At least 40 types of cytokines exist in humans that differ in terms of the cell type that produces them, the cell type that responds to them, and the changes they produce. Interferon is one type of cytokine.

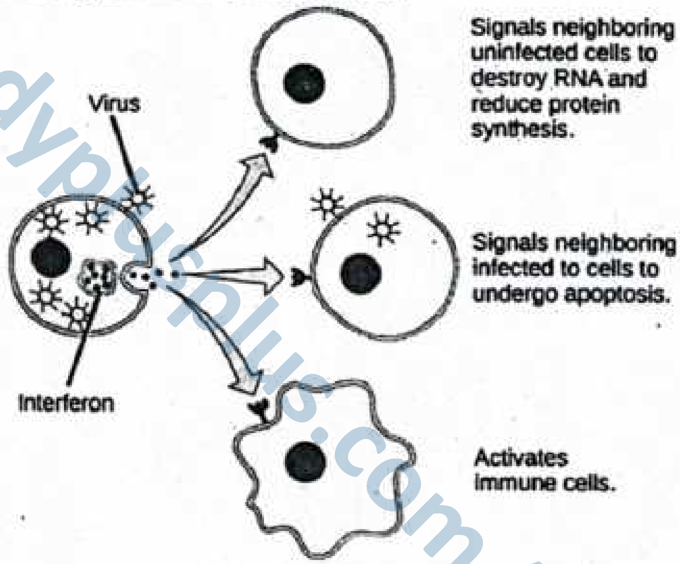

Figure: Action of interferon to control virus

SLO [B-12-1-08]
State the events of the inflammatory response as a generalized, nonspecific defence.

INFLAMMATORY RESPONSE

The inflammatory response is one of the key components of the body's second line of defence. It is a nonspecific reaction that occurs whenever body tissues are injured by physical trauma, chemical irritation, or infection by pathogens. The purpose of inflammation is to isolate the harmful agent, prevent further tissue damage, and promote healing.

1. **Recognition of Injury or Infection:** The process begins when damaged cells, invading microbes, or activated immune cells release chemical signals such as histamine, prostaglandins, and cytokines. These molecules alert the body to the presence of injury or infection and initiate the cascade of events that characterize inflammation.
2. ***Vasodilation and Increased Blood Flow:*** Histamine and related mediators cause the arterioles in the affected area to dilate. As a result, more blood flows into the injured tissue. This produces two of the classic signs of inflammation *redness* (rubor) and *heat* (calor). The increased blood flow brings immune cells and nutrients to the site of injury.

3. **Increased Vascular Permeability:** The capillary walls in the inflamed region become more permeable, allowing plasma proteins, antibodies, and clotting factors to leak into the tissue. This fluid accumulation causes *swelling* (tumor) and exerts pressure on surrounding nerves, resulting in *pain* (dolor).
4. **Migration of Leukocytes (Phagocytes):** Chemical signals (chemokines) attract white blood cells, mainly neutrophils and monocytes, to the site of injury. Neutrophils arrive first, squeezing through the capillary walls in a process known as diapedesis. These cells engulf and destroy microbes through phagocytosis. Monocytes follow later, differentiating into macrophages that provide long-lasting defence and tissue cleanup.
5. **Elimination of Pathogens and Tissue Repair:** As phagocytes destroy invading pathogens, platelets and clotting proteins work to form barriers, preventing the spread of infection. Growth factors released by macrophages stimulate fibroblasts and other cells to repair the damaged tissue. Eventually, swelling subsides and normal tissue function is restored.
6. **The Four Cardinal Signs of Inflammation:** The inflammatory response is typically recognized by four cardinal signs: redness, heat, swelling, and pain. Sometimes a fifth sign, *loss of function*, is also included, reflecting the temporary impairment of the affected tissue during the response.

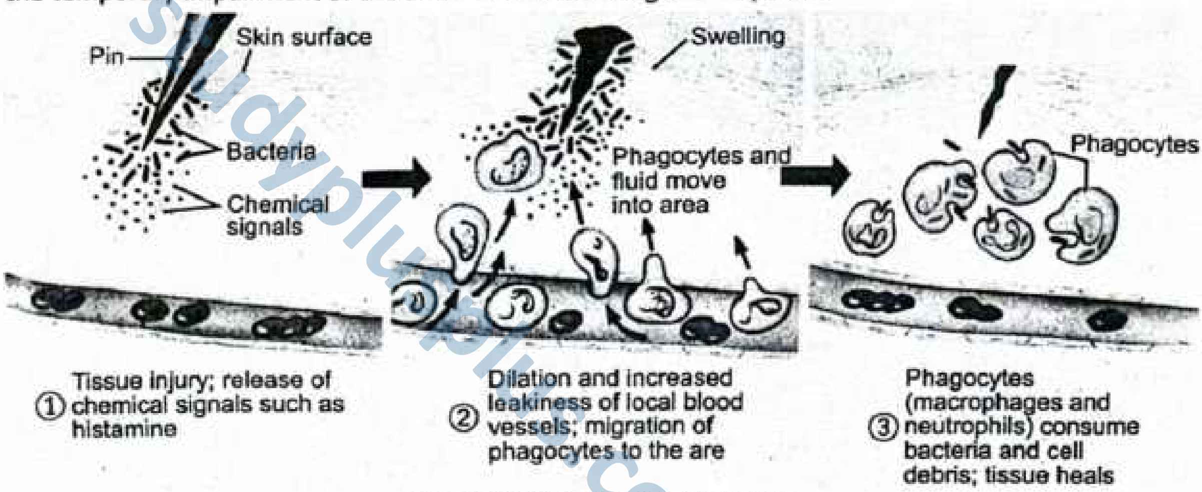

Figure: The Inflammatory Response

SLO [B-12-1-09]
Outline the release of pyrogens by microbes and their effect on the hypothalamus to boost the body's temperature.

PYROGENS AND FEVER: ELEVATING BODY TEMPERATURE

Fever is a **systemic, nonspecific defence mechanism** in which the body raises its core temperature to inhibit microbial growth and enhance immune function. This response is triggered by the release of **pyrogens**, substances that affect the hypothalamic thermoregulatory center. **Fever,** or **pyrexia,** refers to an elevation in body temperature above the normal physiological range, typically exceeding **37°C (98.6°F). It is a non-specific immune response** that plays a supportive role in controlling microbial infections.

1. **Release of Pyrogens by Microbes**

Fever is typically triggered by the presence of infectious agents such as bacteria, viruses, or their by-products. These invading microorganisms often release substances known as **exogenous pyrogens** (e.g., bacterial endotoxins), which stimulate immune cells, particularly **macrophages and other white blood cells,** to produce **endogenous pyrogens**. These endogenous pyrogens include cytokines such as:

- Interleukin-1 (IL-1)
- Interleukin-6 (IL-6)
- Tumour Necrosis Factor-alpha (TNF-α)

2. **Action on the Hypothalamus**

Endogenous pyrogens travel through the bloodstream to the **hypothalamus**, the brain's temperature-regulating center. They stimulate the hypothalamus to increase the set point for body temperature above the normal 37 °C.

3. **Physiological Changes Leading to Fever**

To achieve the new set point, the body initiates heat-conserving and heat-generating mechanisms:

- **Vasoconstriction** of skin arterioles reduces heat loss through the skin, causing chills and pale skin.
- **Shivering** and increased skeletal muscle activity generate additional heat.
- **Increased metabolic rate** mediated by thyroid hormones contributes to internal heat production.

4. **Purpose of Fever**

The elevated body temperature inhibits the replication of many pathogens, enhances **phagocytosis by neutrophils and macrophages,** and accelerates **immune reactions,** such as antibody production. Fever thus represents a critical **nonspecific defence mechanism** that complements other components of innate immunity.

SLO [B-12-1-10]
List the ways that fever affects microbes.

PROTECTIVE ROLES OF FEVER

Although fever is energetically costly, it provides several advantages during infection:

- **Enhancement of Immune Function**: Elevated body temperature increases the activity of **phagocytic leukocytes,** particularly **neutrophils and macrophages**, enhancing the engulfment and destruction of pathogens.
- **Iron Sequestration**: Fever-associated cytokines reduce the availability of **free iron** in the blood, as many bacteria require iron for replication. The reduced iron levels, especially at temperatures around **38–39°C,** impair microbial proliferation.
- **Stimulation of Interferon Production**: Higher temperatures promote the synthesis of **interferons,** which increase resistance of surrounding uninfected cells to viral entry and replication.
- **Speeds Up Tissue Repair:** The increased metabolic rate during fever accelerates immune responses and tissue repair mechanisms.
- **Direct Inhibition of Microbes:**
  - **Enveloped viruses**, which are more heat-sensitive, may be structurally damaged at elevated temperatures.
  - Fever can **slow viral replication** by disrupting optimal conditions required for viral enzyme activity and genome synthesis.

Thus, fever acts as a **systemic response** that enhances immune efficiency and suppresses microbial survival and replication.

Risks Associated with High Fever

Despite its protective role, **excessively high fever** can be detrimental. When body temperature rises beyond **40°C (104°F)**, enzymatic and metabolic processes may become impaired, leading to:

- Protein denaturation
- Disruption of cellular functions
- Dehydration
- Neurological dysfunction

In extreme cases, uncontrolled fever may result in **systemic collapse**, particularly in vulnerable individuals such as infants, the elderly, or patients with underlying health conditions.

Medical Management

To mitigate the harmful effects of high-grade fever, ***antipyretic drugs*** *such as* **paracetamol (acetaminophen)** and **ibuprofen** are commonly administered. These medications inhibit **cyclooxygenase enzymes (COX-1 and COX-2)** involved in prostaglandin synthesis in the hypothalamus, thereby lowering the temperature set point and restoring normal thermoregulation.

Justification for the Use of Antipyretic Drugs

Although fever is a non-specific defence mechanism that enhances immune activity and inhibits pathogen growth, excessively high or prolonged fever can be harmful. Physicians prescribe **antipyretic drugs** to prevent complications such as dehydration, metabolic disruption, or neurological damage.

Antipyretics work by **inhibiting prostaglandin production** in the hypothalamus, blocking the rise in temperature set point. This helps restore normal body temperature without completely suppressing the immune response, ensuring both **patient safety** and **effective recovery**.

SLO [B-12-1-11]
Define the specific immune system as providing specific defence and acting as the most powerful means of resisting infection.

THE SPECIFIC IMMUNE SYSTEM

The **specific immune system**, also referred to as the **adaptive or acquired immune system**, is a highly specialized component of the body's defences that provides **targeted and powerful protection** against pathogens. Unlike the innate immune system, which offers **immediate but generalized** defence, the specific immune system recognizes and responds to **particular antigens**—unique molecular structures present on the surface of microbes, abnormal cells, or foreign substances.

This system operates through two main mechanisms: **humoral immunity**, mediated by B lymphocytes that produce **antibodies** against extracellular pathogens, and **cell-mediated immunity**, mediated by T lymphocytes that destroy infected or abnormal cells. The hallmark of the specific immune system is its **ability to remember previous encounters** with pathogens. This immunological memory allows for a **faster, stronger, and more precise response** upon subsequent exposures, making it the most effective means of resisting infection.

In summary, the specific immune system provides **specificity, memory, and amplification**, enabling the body to selectively target and eliminate pathogens, while also establishing long-term protection against diseases. Its precision and adaptability make it the **most powerful line of defence** in the human immune system.

DO YOU KNOW?

Do you remember, the antigens A and B on red blood cells, you learned in ABO blood group system? and other pathogen have specific antigens on their surface. Since, the third line of defence respond against particular infections and acts as the most powerful means of resisting infections therefore, it is also called specific defence. Its response can be of two types: Humoral immune response and cell-mediated immune response. These immune responses are particularly carried out by two components: B-lymphocytes or B cells and T-lymphocytes or T cells. However, third line of defence also involves role of monocytes (macrophages) that participate in activation of these lymphocytes.

SLO [B-12-1-01]
Identify monocytes, T cells, and B-cells as components of the immune system.

CELLULAR COMPONENTS OF THE IMMUNE SYSTEM

The human immune system relies on a diverse population of white blood cells (leukocytes) that work together to detect, neutralize, and eliminate pathogens. Among these, **monocytes, T lymphocytes (T cells), and B lymphocytes (B cells)** play central roles in both innate and adaptive immunity.

1. Monocytes

Monocytes are **large, mononuclear white blood cells** produced in the bone marrow and circulating in the blood. They *are precursors* to **macrophages** and **dendritic cells**, which migrate into tissues to perform phagocytosis, antigen presentation, and secretion of signalling molecules (cytokines). By linking **innate immunity** to **adaptive immunity**, monocytes serve as essential mediators of the body's defence against infection.

2. T Lymphocytes (T Cells)

T cells are a critical component of the **cell-mediated adaptive immune system**. They mature in the thymus and differentiate into several subtypes, each with specialized functions:

- **Helper T cells ($CD4^+$):** Activate B cells and cytotoxic T cells, and coordinate the immune response through cytokine secretion.
- **Cytotoxic T cells ($CD8^+$):** Directly destroy virus-infected or abnormal cells by inducing apoptosis.
- **Regulatory T cells:** Suppress overactive immune responses to maintain immune tolerance and prevent autoimmunity.

3. B Lymphocytes (B Cells)

B cells are the main effectors of **humoral immunity**. They mature in the bone marrow and, upon encountering a specific antigen, differentiate into **plasma cells** that secrete **antibodies.** These antibodies bind to extracellular pathogens or toxins, neutralizing them or marking them for destruction by other immune cells. Memory B cells are also formed, providing **long-term immunological memory** for rapid responses upon subsequent exposure to the same pathogen.

Together, **monocytes, T cells, and B cells** form the backbone of the human immune system, integrating innate and adaptive mechanisms to provide both **immediate and long-term protection** against infections.

SLO [B-12-1-13]
State inborn and acquired immunity as the two basic types of immunity.

TYPES OF IMMUNITY

The human body defends itself against a wide variety of pathogens, including bacteria, viruses, fungi, and parasites, through two fundamental types of immunity: **inborn (innate) immunity** and **acquired (adaptive) immunity**. These two systems function together to provide comprehensive protection, yet they differ in speed, specificity, and the ability to remember previous encounters with pathogens.

1. Inborn (*Innate*) Immunity

Inborn or **innate immunity** is present from birth and forms the body's **first line of defence** against infection. It provides **rapid, generalized protection** against a wide array of harmful agents. Innate immunity is **nonspecific,** meaning it does not distinguish between different types of pathogens. Key components include **physical barriers** like the skin and mucous membranes, which prevent microbial entry, and **chemical barriers** such as gastric acid in the stomach, lysozymes in saliva, and antimicrobial fatty acids in sebum. Cellular elements of innate immunity include **phagocytic cells** like neutrophils and macrophages, which engulf and destroy microbes, as well as **natural killer (NK) cells**, which target virus-infected and cancerous cells. Inborn immunity is critical for **immediate response**, acting within minutes to hours after infection, but it **lacks immunological memory,** meaning that repeated exposure to the same pathogen elicits a similar response each time.

2. Acquired (Adaptive) Immunity

Acquired or **adaptive immunity** develops gradually after birth in response to exposure to specific pathogens. It is **highly specific,** targeting unique antigens present on microbes or foreign substances. The two main branches of adaptive immunity are **humoral immunity**, mediated by B lymphocytes that produce antibodies, and **cell-mediated immunity**, mediated by T lymphocytes that destroy infected or abnormal cells. A defining feature of acquired immunity is **immunological memory**: once the body encounters a pathogen, it "remembers" it, allowing for a faster and stronger response upon subsequent exposures. Acquired immunity can be **naturally acquired**, as through infection, or **artificially acquired**, as through vaccination. Vaccines stimulate the immune system to produce memory B and T cells without causing the disease, *providing* ***long-term protection***.

3. Complementary Roles

Inborn and acquired immunity are not independent; they work synergistically. Innate immunity provides **immediate defence,** containing the infection and signaling the adaptive system. Components of the innate system, such as macrophages and dendritic cells, also **present antigens** to T lymphocytes, effectively bridging innate and adaptive responses. Acquired immunity, in turn, provides **specificity, amplification, and long-term protection**, ensuring that subsequent infections are neutralized more efficiently. Together, these two types of immunity form a **comprehensive defence network**, protecting the body from the vast array of pathogenic challenges it encounters throughout life.

SCIENCE TITBITS

In 1717 Mary Montagu, the wife of an English ambassador to the Ottoman Empire, observed local women inoculating their children against smallpox. Edward Jenner observed and studied Miss Sarah a milkmaid who had previously caught cowpox and was found to be immune to smallpox.

SLO [B-12-1-14]
Differentiate between active and passive immunity as the two types of acquired immunity.

TYPES OF ACQUIRED IMMUNITY: ACTIVE VS. PASSIVE

Acquired immunity, also called **adaptive immunity**, can be further classified into **active** and **passive immunity**, depending on whether the immune system itself produces the protective response or receives it from another source. Both forms provide **specific defence against pathogens**, but they differ in onset, duration, and mechanism.

1. Active Immunity

Active immunity arises when an individual's immune system **actively produces antibodies and memory cells** in response to exposure to a specific antigen. This can occur naturally, such as after **infection by a pathogen**, or artificially, such as through **vaccination**. In active immunity, the immune system is directly engaged: B lymphocytes are stimulated to produce antibodies, and T lymphocytes are activated to recognize and destroy infected cells. The primary advantage of active immunity is that it **generates long-term or even lifelong protection**, due to the formation of **memory B and T cells**. However, the development of full immunity requires time, often several days to weeks, before antibody levels peak.

Examples of Active Immunity:

- **Natural:** Recovery from diseases like measles or chickenpox.
- **Artificial:** Immunization with vaccines such as polio, hepatitis B, or tetanus toxoid.

2. Passive Immunity

Passive immunity occurs when an individual **receives preformed antibodies** from another source rather than producing them. This provides **immediate protection**, but it is **short-lived**, as the transferred antibodies are gradually degraded and no memory cells are formed. Passive immunity can be naturally acquired, such as through **maternal antibodies transferred via the placenta (IgG) or breast milk (IgA)**, or artificially, such as through **administration of antiserum or monoclonal antibodies** to treat infections or prevent disease.

Examples of Passive Immunity:

- **Natural:** Antibodies received by a newborn from the mother.
- **Artificial:** Injection of antivenom for snake bites or immunoglobulins for hepatitis exposure.

▶ Comparison and Significance

Active immunity is **slow to develop but long-lasting**, providing durable protection, while passive immunity is **rapid but temporary**, useful in immediate threat situations or for short-term protection. Together, these two forms of acquired immunity allow the immune system to respond effectively to both **long-term prevention and acute exposure** scenarios.

Feature	Active Immunity	Passive Immunity
Definition	Immunity developed by the body's own immune system after exposure to an antigen.	Immunity provided by **transfer of ready -made antibodies** from another source.
Source of Antibodies	Produced by the individual's own B lymphocytes.	Received from an external source (another individual or animal).
Development Time	Takes time to develop (days to weeks).	Provides immediate protection.

Duration	Long-lasting, sometimes lifelong (due to memory cell formation).	Short-term (antibodies degrade over time).
Examples	**Natural:** After recovering from measles **Artificial:** Vaccination	**Natural:** Antibodies passed from mother to baby via placenta or breast milk **Artificial:** Injection of antiserum (e.g. for rabies or snake venom)
Memory Cells Formed?	Yes	No

PRIMARY AND SECONDARY IMMUNE RESPONSES

The immune system responds to pathogens in two phases:

- **Primary Immune Response:** This occurs upon the first exposure to a specific antigen. It is relatively slow and weak, as the immune system takes time to recognize the antigen and activate B and T cells. Memory cells are generated during this phase.
- **Secondary Immune Response:** Upon subsequent exposures to the same antigen, memory B and T cells facilitate a much faster and more robust immune response. This forms the basis of long-term immunity and the effectiveness of vaccines.

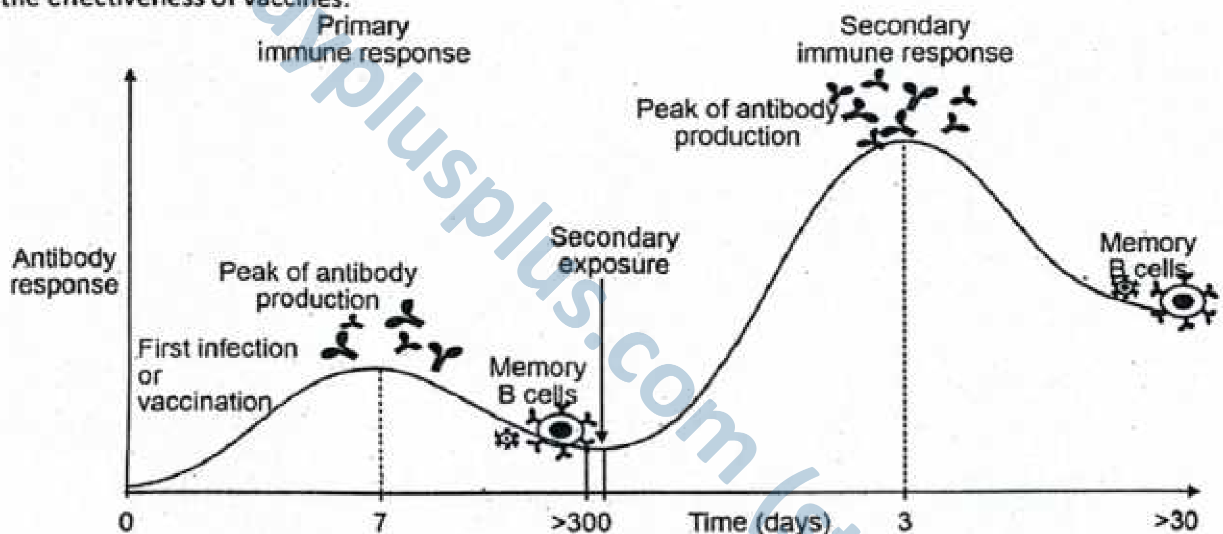

Figure: Primary vs Secondary Immune response

SLO [B-12-1-15]
Describe the role of T-cells in cell-mediated immunity.

ROLE OF T CELLS IN THIRD LINE OF DEFENCE (CELL MEDIATED IMMUNE RESPONSE)

T lymphocytes (T cells) originate from stem cells in the bone marrow and mature in the **thymus**, where they become **immunocompetent**, i.e., capable of recognizing specific antigens. The response involving T cells is called the **cell-mediated immune response**.

Activation of T Cells

During infection:

1. **Macrophages or dendritic cells** present pathogen-derived antigens on their surface using MHC molecules.
2. These antigen-presenting cells secrete **interleukin-1**, which recruits **helper T cells ($CD4^+$)** to the site.
3. Each helper T cell has a unique **T cell receptor (TCR)** specific to a particular antigen. Upon antigen recognition, IL-1 stimulates the helper T cells to secrete **interleukin-2**.

4. IL-2 promotes:
   - Clonal expansion of helper T cells,
   - Activation and proliferation of **cytotoxic T cells ($CD8^+$)**,
   - Activation of **B cells** (involved in humoral immunity).

This highly regulated interaction initiates **cell-mediated immunity**, enabling the immune system to eliminate infected or abnormal host cells.

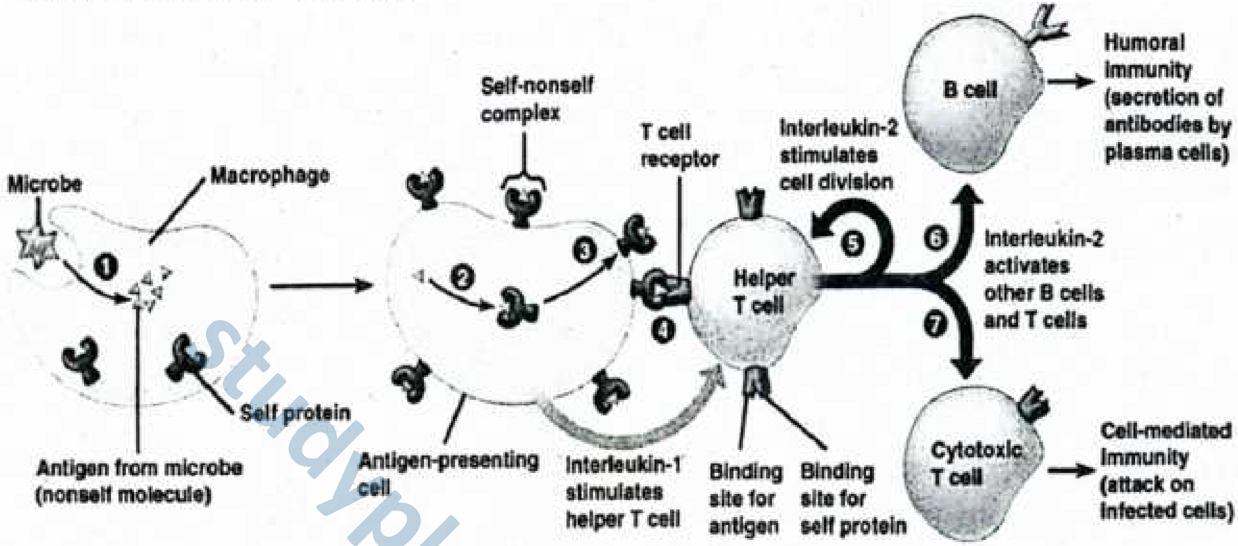

Figure: Activation of cell mediated immune response

Types of T Cells and Their Functions

Upon activation, T cells differentiate into four main types, each with a specific function:

T Cell Type	Function
Cytotoxic T Cells ($CD8^+$)	Recognize and kill infected or abnormal cells by releasing: • **Perforins**, which form pores in target cell membranes. • **Cytotoxins**, which induce apoptosis or DNA damage.
Helper T Cells ($CD4^+$)	Secrete **interleukin-2** to activate other T cells and stimulate B cell proliferation and antibody production. They coordinate the overall immune response.
Suppressor (Regulatory) T Cells	Inhibit T cell activity after infection is resolved by secreting regulatory proteins. They help prevent autoimmune reactions.
Memory T Cells	Persist long-term after infection. On re-exposure to the same antigen, they rapidly differentiate into active T cells, ensuring a faster and stronger response.

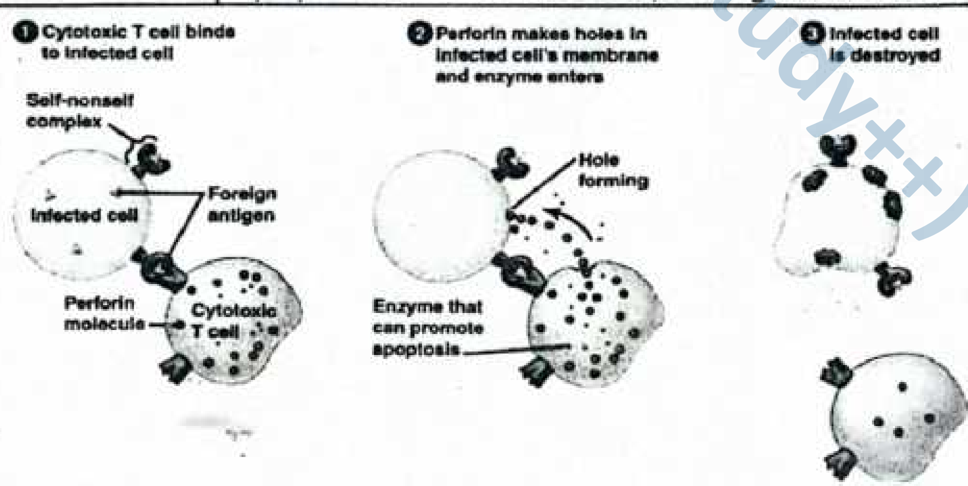

Figure: Mode of action of cytotoxic T cells

SLO [B-12-1-16]
Describe the role of B-cells in antibody-mediated immunity.

ROLE OF B CELLS IN THIRD LINE OF DEFENCE: HUMORAL/ ANTIBODY MEDIATED IMMUNE RESPONSE

B-lymphocytes (B-cells) are central to the body's **antibody-mediated (humoral) immune response**, which targets pathogens circulating in body fluids such as blood and lymph. Unlike T-cells, which primarily act on infected cells, B-cells are specialized for producing antibodies that neutralize extracellular microbes and their toxins.

Role of B-cells in antibody-mediated immunity

1. Activation of B-Cells

B-cells originate in the bone marrow, where they mature and acquire specific receptors (B-cell receptors, BCRs) that recognize unique antigens. Upon encountering their specific antigen, B-cells become activated. This activation often requires assistance from **helper T-cells (CD4+ T-cells)**, which release cytokines to stimulate B-cell proliferation and differentiation.

2. Clonal Expansion and Differentiation

Once activated, B-cells undergo **clonal expansion**, producing a large population of identical cells. These cells differentiate into:

- **Plasma Cells**: Short-lived effector cells that secrete large quantities of antibodies specific to the invading antigen.
- **Memory B-Cells**: Long-lived cells that remain in the body, ensuring a faster and stronger antibody response upon re-exposure to the same pathogen.

3. Antibody Production and Action

The antibodies secreted by plasma cells circulate in the bloodstream and lymph, where they bind specifically to the antigens of pathogens. Their defensive actions include:

- **Neutralization**: Antibodies block the attachment of microbes or toxins to host cells.
- **Opsonization**: Antibodies coat microbes, marking them for phagocytosis by macrophages and neutrophils.
- **Complement Activation**: The antibody-antigen complex activates the complement system, leading to microbial lysis.
- **Agglutination and Precipitation**: Antibodies cause microbes or toxins to clump together, making them easier targets for immune cells.

4. Immunological Memory

The formation of memory B-cells ensures **long-term immunity**. This is the basis of vaccination, where exposure to harmless forms of antigens trains the immune system to respond rapidly upon later infection.

In summary, B-cells play a vital role in antibody-mediated immunity by recognizing antigens, differentiating into plasma cells and memory cells, producing antibodies, and providing long-lasting protection against reinfection.

Development *and* Characteristics of B Cells

B cells originate and mature *in the bone marrow*. Each B cell displays unique membrane-bound receptors known as **B cell receptors (BCRs)**, which are specific to a particular antigen. These BCRs enable B cells to recognize and bind to antigens with high specificity. The human immune system contains millions of B cells, each capable of recognizing a different antigen.

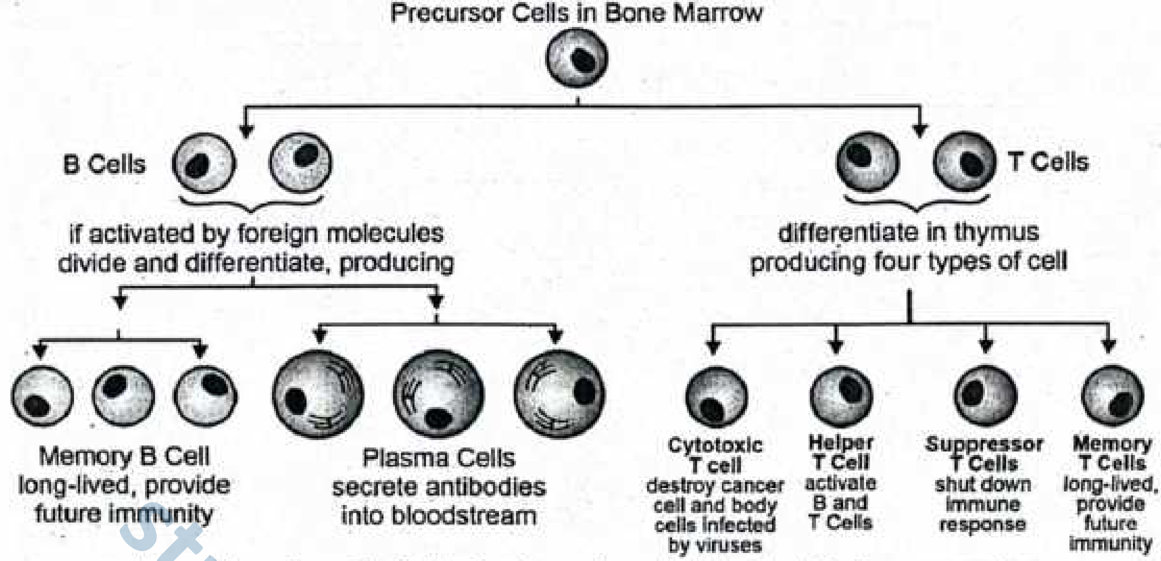

Figure: The major cells of Adaptive Immunity and their roles in the immune system

SLO [B-12-1-17]
Discuss the role of T -cells and B -cells in transplant rejections.

ROLE OF T-CELLS AND B-CELLS IN TRANSPLANT REJECTIONS

When an organ or tissue is transplanted from one individual to another, the recipient's immune system often recognizes the graft as "foreign" because of genetic differences in surface proteins, particularly the **major histocompatibility complex (MHC) molecules** (also known as human leukocyte antigens, HLA, in humans). This immune recognition leads to a complex response in which both **T-cells** and **B-cells** play critical roles in graft rejection.

1. Role of T-Cells

T-lymphocytes are the primary mediators of transplant rejection.

- **Cytotoxic T-Cells ($CD8^+$ T-cells):** These cells directly recognize foreign MHC molecules on the donor's transplanted tissue. Once activated, they attack and destroy the graft cells by releasing perforins and granzymes, causing cell lysis and apoptosis.
- **Helper T-Cells ($CD4^+$ T-cells):** These cells recognize antigens presented by antigen-presenting cells (APCs) and release cytokines that amplify the immune response. They recruit macrophages, activate cytotoxic T-cells, and stimulate B-cell activity.
- **Result:** T-cell-mediated destruction is the main mechanism behind **acute cellular rejection**, often occurring days to weeks after transplantation.

2. Role of B-Cells

B-lymphocytes contribute mainly through **antibody-mediated rejection (AMR).**

- **Antibody Production:** Once activated, B-cells differentiate into plasma cells and secrete antibodies against donor antigens, particularly the foreign MHC proteins.
- **Antibody Action:** These antibodies bind to graft endothelial cells, activating the **complement system.** This leads to inflammation, vascular damage, and thrombosis, ultimately impairing blood supply to the graft.
- **Result:** Antibody-mediated rejection is a major factor in **hyperacute rejection** (within minutes to hours) and **chronic rejection** (months to years after transplantation).

3. Combined Response

Since both T-cells and B-cells are activated, transplant rejection is the result of a **coordinated immune attack:**

- T-cells destroy graft cells directly and recruit more immune cells.
- B-cells produce antibodies that damage graft vasculature and enhance inflammation.

4. **Clinical Implication**

- Understanding the role of T-cells and B-cells in transplant rejection forms the basis of using **immunosuppressive drugs** (e.g., cyclosporine, corticosteroids, monoclonal antibodies) in transplant patients to dampen the immune response and promote graft survival.

SLO [B-12-1-18]
Evaluate the discovery of monoclonal antibodies and justify how this accomplishment revolutionized many aspects of biological research.

DISCOVERY AND SIGNIFICANCE OF MONOCLONAL ANTIBODIES

Discovery of Monoclonal Antibodies

Monoclonal antibodies (mAbs) were first developed in **1975** by **Georges Köhler and César Milstein**, who were later awarded the **Nobel Prize in Physiology or Medicine (1984)** for this groundbreaking achievement. They developed a method called **hybridoma technology**, which involves fusing a specific antibody-producing **B-cell** with a rapidly dividing **myeloma (cancer) cell**. The resulting hybridoma can:

1. Grow indefinitely in laboratory culture.
2. Produce large quantities of a **single, identical (monoclonal) antibody** that targets one specific antigen.

This discovery was revolutionary because it overcame the limitations of traditional polyclonal antibodies, which are heterogeneous and less specific.

Why Monoclonal Antibodies Revolutionized Biological Research

The discovery of monoclonal antibodies brought an unprecedented transformation to biological research, diagnostics, and medicine. Their unique ability to recognize a single epitope with high precision has made them one of the most powerful tools in modern science.

1. **Unprecedented Specificity**

Monoclonal antibodies exhibit remarkable precision by binding to a single epitope on an antigen. This high specificity allows researchers to track, measure, and manipulate particular proteins, cells, or molecules with minimal cross-reactivity. Such accuracy has made them indispensable in both experimental and clinical applications.

2. **Biomedical Research**

In the field of research, monoclonal antibodies are widely employed to identify and isolate specific cell types, such as T-cells, B-cells, or stem cells. They are also fundamental in advanced molecular biology techniques. For instance, in immunohistochemistry, flow cytometry, and western blotting, monoclonal antibodies serve as essential tools for detecting and analyzing proteins within tissues and cells.

3. **Medical Diagnostics**

One of the greatest impacts of monoclonal antibodies has been in medical diagnostics. They form the foundation of diagnostic kits such as **ELISA tests**, which are used for detecting viral infections like HIV and hepatitis. More recently, monoclonal antibodies have been central to the development of **rapid antigen tests** for diseases such as COVID-19. By enabling accurate and early disease detection, these tools significantly improve patient outcomes.

4. **Therapeutics and Clinical Use**

Monoclonal antibodies have revolutionized treatment strategies through their role as targeted therapies. In oncology, drugs like **trastuzumab** (for breast cancer) and **rituximab** (for lymphoma) specifically target cancer cells, reducing collateral damage to healthy tissue. They are also invaluable in managing autoimmune diseases, such as rheumatoid arthritis and Crohn's disease, by neutralizing inflammatory molecules. Additionally, monoclonal antibodies are now used as antiviral agents, such as therapies developed against **SARS-CoV-2** during the COVID-19 pandemic.

5. **Safer Alternatives**

Compared to whole serum therapies, monoclonal antibodies offer a safer and more reliable option. Their production is standardized and reproducible, ensuring consistency between batches. Furthermore, they are less likely to cause allergic or adverse immune reactions, making them a preferred therapeutic choice.

6. **Industrial and Agricultural Applications**

Beyond medicine, monoclonal antibodies have proven valuable in industry and agriculture. In food safety, they are employed to detect toxins, pathogens, and contaminants, ensuring consumer protection. In agriculture, they are used to diagnose plant diseases and contribute to the development of disease-resistant crops, supporting sustainable farming practices.

Evaluation

The discovery of monoclonal antibodies marked a **turning point in immunology, medicine, and biotechnology**. Their production provided researchers with a reliable and renewable source of highly specific antibodies, transforming not only laboratory research but also diagnostics, therapeutics, and industrial applications. The hybridoma technique remains one of the most important innovations in biomedical science, laying the foundation for modern targeted therapies and precision medicine.

The development of monoclonal antibodies by Köhler and Milstein revolutionized biology because it provided a precise, renewable tool for detecting, diagnosing, and treating diseases. Their impact spans from research laboratories to hospitals, making them one of the most powerful tools in modern science.

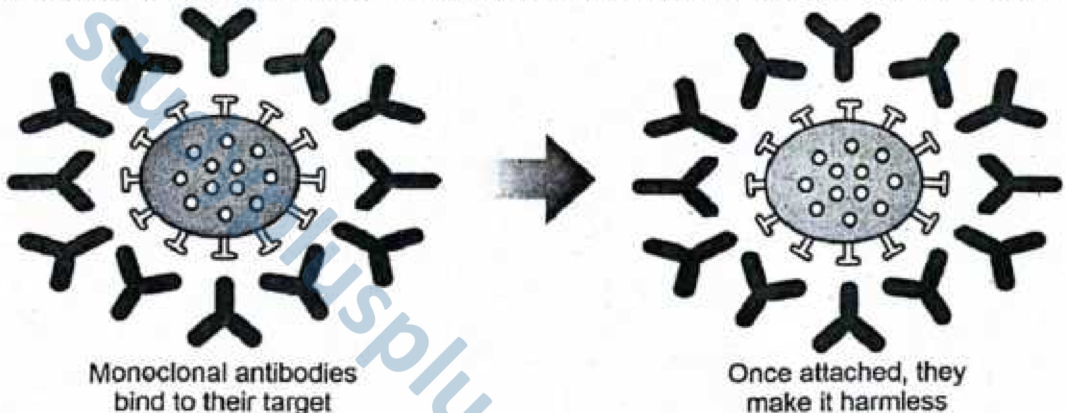

Figure: Action of mAbs onCOVID-19 virus to control the disease

SLO [B-12-1-19]
Identify the process of vaccination as a means to develop active acquired immunity.

VACCINATION AND ACTIVE ACQUIRED IMMUNITY

Vaccination is a scientific method used to stimulate the body's immune system to develop **active acquired immunity** without causing disease. It is one of the most important achievements in medical science, responsible for controlling and even eradicating deadly infectious diseases worldwide.

The Process of Vaccination

In vaccination, a harmless form of a pathogen or its components is deliberately introduced into the body. These may include:

1. **Antigen Selection**: Scientists choose a *safe* form of the pathogen (the antigen) that can train the immune system without causing disease. Common formats include:
   - Inactivated (killed) pathogen
   - Attenuated (live, weakened) pathogen
   - Toxoid (inactivated bacterial toxin)
   - Subunit/Recombinant antigen (purified protein or polysaccharide; may be conjugated to a carrier protein)
2. **Formulation with Adjuvants and Stabilizers**: The antigen is formulated with **adjuvants** (e.g., aluminum salts) to enhance immune recognition, plus **stabilizers/preservatives** to maintain potency and safety during storage.
3. **Administration (Exposure)**: The vaccine is **delivered** (typically intramuscular, subcutaneous, intradermal, or intranasal). This introduces the antigen to immune cells in skin/muscle and local lymphatics.

4. **Innate Immune Sensing**: **Dendritic cells and macrophages** at the injection site detect the antigen (and adjuvant) via pattern-recognition receptors, leading to **local inflammation**, cytokine release, and **antigen uptake**.
5. **Antigen Processing and Presentation**: Dendritic cells **process** the antigen and migrate to **draining lymph nodes**, presenting antigen fragments on **MHC II (to CD4⁺ T cells)** and, for some vaccines/antigens, on **MHC I (cross-presentation to CD8⁺ T cells)**.
6. **T-Cell Priming (Helper and Cytotoxic)**: **CD4⁺ helper T cells** are activated and differentiate (e.g., Th1/Th2/Th17), secreting cytokines that guide B-cell responses. With certain vaccine types (notably attenuated), **CD8⁺ cytotoxic T cells** may also be primed to kill infected cells in future exposures.
7. **B-Cell Activation and Clonal Expansion**: **B cells** that bind the antigen via their BCR internalize and present it to helper T cells. Receiving **T-cell help** (CD40–CD40L plus cytokines), B cells **proliferate** and form **germinal centers**.
8. **Differentiation: Plasma Cells and Antibody Secretion**
   - Activated B cells differentiate into **plasma cells**, producing high levels of **antibodies** (initially IgM, then class-switched IgG/IgA/IgE depending on vaccine and route).
   - Antibodies **neutralize toxins/viruses**, **opsonize** pathogens for phagocytosis, **activate complement**, and **agglutinate** antigens.
9. **Affinity Maturation and Class Switching**: In germinal centers, B cells undergo **somatic hypermutation** and **class switching**, yielding **high-affinity antibodies** optimized for protective function and tissue distribution.
10. **Memory Formation**: A subset becomes **long-lived memory B cells** and **memory T cells** (and, for some vaccines, **long-lived plasma cells** in bone marrow), establishing **active acquired immunity**.
11. **Booster Doses (If Required)**: **Boosters** re-expose the immune system to the antigen, expanding memory pools and increasing antibody **titer and affinity**, prolonging protection.
12. **Protective Response on Real Exposure**: Upon encountering the actual pathogen, **memory cells respond rapidly** with a **faster, stronger (anamnestic) response**, preventing or markedly reducing disease.

Where Vaccine Types Fit in

- **Inactivated and Subunit/Recombinant Vaccines**: These vaccines primarily stimulate the **humoral immune response**, leading to the production of protective antibodies. Since they cannot replicate, they often require **adjuvants** to enhance immunogenicity and **booster doses** to maintain long-term protection.
- **Toxoid Vaccines**: These are based on **inactivated bacterial toxins**. They train the immune system to produce **antitoxin antibodies** that neutralize the harmful effects of toxins rather than the pathogen itself. Examples include **tetanus** and **diphtheria** vaccines.
- **Attenuated (Live, Weakened) Vaccines**: Because these mimic natural infections, they activate both **humoral (antibody-mediated)** and **cell-mediated** immune responses. This dual stimulation generates stronger and **long-lasting memory**, often providing protection with fewer doses. Examples include **measles, mumps, and rubella (MMR)** vaccines.

Development of Active Acquired Immunity

The immunity that results from vaccination is termed **active acquired immunity**. This process involves:

1. **Active Production of Antibodies** – The immune system is stimulated by the harmless antigen introduced through the vaccine, leading to the **body's own production of antibodies** against the pathogen.
2. **Formation of Memory Cells** – Along with antibodies, the immune response generates **memory B and T cells**. These cells "remember" the antigen and ensure a **faster, stronger, and more effective response** if the pathogen invades again.
3. **Long-Term Protection** – Because of the establishment of immunological memory, the protection gained is **long-lasting**, and in some cases (such as measles or hepatitis B vaccination), it can be **lifelong**, depending on the vaccine type.

Importance of Vaccination

- Prevents serious infectious diseases such as smallpox, measles, polio, and hepatitis B.
- Reduces the spread of pathogens in the community (herd immunity).

- Protects vulnerable individuals, such as infants, elderly, and immunocompromised patients.
- Contributes to global disease eradication campaigns (e.g., eradication of smallpox, near elimination of polio).

SLOs Based (MCQs)

LINES OF DEFENCE

▶ **First Line of Defence**

1. Which of the following is a component of the first line of defence?
A. Natural killer cells B. Saliva
C. Interleukin-2 D. Cytotoxic T cells

2. Lysozyme, an enzyme that digests bacterial cell walls, is found in:
A. Red blood cells B. Lymph nodes
C. Sweat and tears D. Bone marrow

3. Which structure prevents the entry of pathogens into the lower respiratory tract?
A. Epiglottis B. Nasal hair
C. Cilia in trachea D. Vocal cords

4. Which of the following is not part of the physical barriers in the first line of defence?
A. Mucous membranes B. Skin
C. Stomach acid D. Tears

5. The acidity of the stomach primarily serves to:
A. Digest proteins B. Absorb nutrients
C. Destroy pathogens D. Neutralize enzymes

6. Which cells are directly involved in the third line but not the first line of defence?
A. Skin cells B. B cells
C. Goblet cells D. Ciliated epithelial cells

7. The first line of defence is:
A. Specific B. Adaptive
C. Immediate and non-specific
D. Delayed and specific

▶ **Second Line of Defence**

8. Which of the following is a phagocytic white blood cell?
A. Basophil B. Monocyte
C. Eosinophil D. Platelet

9. Fever is beneficial to the host because:
A. It increases pathogen growth
B. It slows down immune responses
C. It enhances phagocytic activity and interferon production
D. It suppresses B cell activity

10. Interferons primarily help the body by:
A. Destroying bacteria
B. Inhibiting viral replication
C. Producing antibodies
D. Activating the complement system

11. Which of the following cells release histamine during inflammation?
A. Macrophages
B. Basophils and mast cells
C. Neutrophils D. Lymphocytes

12. Natural killer (NK) cells destroy infected cells by:
A. Producing antibodies
B. Releasing perforin and granzymes
C. Phagocytosis
D. Antigen presentation

13. Which system involves plasma proteins that enhance the immune response?
A. Lymphatic system B. Complement system
C. Endocrine system D. Respiratory system

14. Which is not a feature of inflammation?
A. Redness B. Swelling
C. Memory cell formation D. Heat

15. The function of phagocytes in the second line of defence is to:
A. Produce antigens
B. Kill infected host cells
C. Engulf and digest pathogens
D. Secrete antibodies

16. Which of the following best describes endogenous pyrogens?
A. Toxins from bacteria
B. Interleukin-1 from white blood cells
C. Viral particles
D. Antibodies

17. Why do physicians prescribe antipyretic drugs?
A. To kill viruses directly
B. To enhance inflammation
C. To inhibit prostaglandin production and reduce fever
D. To increase iron availability

▶ **Third Line of Defence**

18. The third line of defence is characterized by:
A. Non-specific responses B. Rapid response
C. Specificity and memory D. Phagocytosis only

19. **What activates helper T cells during an immune response?**
A. Direct contact with bacteria
B. Cytokines from B cells
C. Antigen presentation by macrophages
D. Histamine release

20. **The function of cytotoxic T cells is to:**
A. Produce antibodies
B. Release histamine
C. Destroy infected cells by releasing perforin
D. Activate macrophages

21. **Plasma cells are derived from:**
A. Helper T cells B. Cytotoxic T cells
C. B cells D. Macrophages

22. **The first step in B cell activation is:**
A. Binding with antigen via BCR
B. Secretion of interferons
C. Production of histamine
D. Formation of memory T cells

23. **Which cytokine activates both B cells and cytotoxic T cells?**
A. Interferon-gamma B. Interleukin-1
C. Interleukin-2 D. Perforin

24. **Memory B cells function to:**
A. Digest pathogens
B. Stimulate neutrophils
C. Provide long-term immunity
D. Secrete histamine

25. **The term "antigen" refers to:**
A. A substance that triggers an immune response
B. A type of antibody
C. A type of white blood cell
D. An enzyme in the stomach

26. **What does "MHC" stand for in antigen presentation?**
A. Major Histo Compatibility
B. Minor Histamine Compound
C. Microbial Host Carrier
D. Macrophage Hematopoietic Complex

27. **Which of the following is true about helper T cells?**
A. They destroy pathogens directly
B. They suppress the immune system
C. They activate other immune cells
D. They secrete perforin

28. **Which T cell is responsible for downregulating immune responses?**
A. Cytotoxic T cell B. Helper T cell
C. Suppressor T cell D. Memory T cell

29. **Which immune cell acts as an antigen-presenting cell (APC)?**
A. Basophil B. Platelet
C. Macrophage D. Plasma cell

30. **Antibody production occurs in:**
A. Helper T cells B. Macrophages
C. Plasma cells D. Cytotoxic T cells

Answer Key

1. B	2. C	3. C	4. C	5. C	6. B	7. C	8. B	9. C	10. B	11. B	12. B	13. B	14. C	15. C
16. B	17. C	18. C	19. C	20. C	21. C	22. A	23. C	24. C	25. A	26. A	27. C	28. C	29. C	30. C

Short Questions

Q.1 What is the function of the first line of defence?

Ans. The first line of defence acts as the body's initial protective barrier against pathogens. It consists of both physical and chemical components that prevent harmful microorganisms from entering the body. The skin provides a tough, impermeable layer, while mucous membranes trap and expel microbes through secretions such as mucus and cilia movement. Chemical barriers include enzymes in saliva and tears (lysozyme) that break down bacterial cell walls, and stomach acid that kills most ingested microbes. These defences are non-specific, meaning they protect against a wide range of pathogens without targeting any particular one. Their role is to minimize infection risk before more specialized immune responses are activated.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark; functions – 2 marks

Q.2 What are pyrogens and how do they contribute to fever?

Ans. Pyrogens are fever-inducing substances that reset the body's thermostat located in the hypothalamus. They may originate from pathogens such as bacteria (exogenous pyrogens) or from the host's immune cells like macrophages (endogenous pyrogens). By raising the hypothalamic set point, pyrogens trigger heat conservation responses

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark; contribution to fever – 2 marks

such as vasoconstriction and shivering, as well as heat production through increased metabolism. This results in an elevated body temperature, known as fever. Fever plays a protective role by slowing the growth of pathogens, enhancing the activity of white blood cells, and stimulating processes like phagocytosis and interferon production. Thus, pyrogens contribute to both direct pathogen inhibition and improved immune
25 function.

Q.3 How do phagocytes work in the immune response?

Ans. Phagocytes, including neutrophils and macrophages, are vital cells of the innate immune system that defend against pathogens through the process of phagocytosis. They identify microbes via surface receptors, engulf them into vesicles called phagosomes, and fuse these vesicles with lysosomes containing digestive enzymes. The pathogens are then broken down and destroyed. In addition to eliminating invaders, macrophages can act as antigen-presenting cells by displaying pathogen fragments on their surface, thereby activating lymphocytes in the adaptive immune response. Phagocytes are fast-acting, non-specific defenders that provide an immediate response to infection. Their dual role of direct destruction and communication with the third line of defence makes them central to overall immunity.

MARKING SCHEME (3 Marks)
- Definition – 1 mark; functions – 2 marks

Q.4 What is the role of natural killer cells?

Ans. Natural killer (NK) cells are specialized lymphocytes of the innate immune system that target abnormal body cells. Unlike other lymphocytes, NK cells do not require prior exposure to a pathogen or antigen. They recognize cells that are infected by viruses or have become cancerous, often by detecting low levels of "self" surface markers such as MHC I proteins. Once identified, NK cells release cytotoxic substances, including perforin, which creates pores in the target cell membrane, and granzymes, which trigger programmed cell death (apoptosis). This rapid response prevents the spread of infection and eliminates potentially harmful cells early. NK cells therefore serve as a crucial first barrier against cancer and viral infections.

MARKING SCHEME (3 Marks)
- Definition – 1 mark; functions – 2 marks

Q.5 Why is inflammation considered part of the second line of defence?

Ans. Inflammation is a protective biological response that occurs when tissues are injured or infected, making it a key feature of the body's second line of defence. It is initiated by the release of histamine and other chemicals from damaged cells or mast cells, which increase blood flow and capillary permeability at the affected site. This allows immune cells, antibodies, and proteins like clotting factors to reach the area more effectively. The redness, swelling, heat, and pain associated with inflammation signal immune activity. Inflammation not only helps isolate and destroy pathogens but also clears damaged tissue and initiates repair. By creating localized immune activity, it enhances healing and prevents further spread of infection.

MARKING SCHEME (3 Marks)
- Definition – 1 mark; function – 2 marks

Q.6 How do B cells recognize specific antigens?

Ans. B cells recognize antigens through specialized proteins on their surface called B cell receptors (BCRs). Each BCR is highly specific to a particular antigen, allowing B cells to detect unique foreign molecules such as proteins, polysaccharides, or toxins. When a BCR binds to its matching antigen, the B cell becomes activated. This activation is often strengthened by signals from helper T cells. Activated B cells then differentiate into plasma cells, which produce large amounts of antibodies specific to the antigen, and into memory B cells, which remain in circulation to provide long-term protection. This process links pathogen detection directly to antibody production, forming a vital component of adaptive immunity.

MARKING SCHEME (3 Marks)
- Definition – 1 mark; function – 2 marks

Q.7 What are memory T cells and their role in immunity?

Ans. Memory T cells are a subset of T lymphocytes that persist long after the initial immune response. They are generated following the first exposure to a specific antigen during an infection or vaccination. Unlike naïve T cells, memory T cells are primed for rapid action. If the same antigen re-enters the body, they quickly recognize it and mount a faster and stronger response compared

MARKING SCHEME (3 Marks)
- Definition – 1 mark; function – 2 marks

to the primary exposure. This quick reaction prevents reinfection or reduces the severity of disease. Memory T cells are central to long-lasting immunity and form the biological basis for vaccination, which trains the immune system to respond effectively to future infections.

Q.8 Differentiate between helper T cells and cytotoxic T cells.

Ans. Helper T cells ($CD4^+$) and cytotoxic T cells ($CD8^+$) are distinct subtypes of T lymphocytes that coordinate different immune functions. Helper T cells do not kill pathogens directly but secrete cytokines such as interleukin-2, which stimulate the activation and proliferation of B cells, cytotoxic T cells, and macrophages. They are essential in bridging innate and adaptive immunity. Cytotoxic T cells, on the other hand, directly destroy infected or abnormal body cells. They release perforin, which forms pores in the target cell membrane, and granzymes, which induce apoptosis. Together, these two T cell types ensure effective immune regulation—helper T cells coordinate responses, while cytotoxic T cells eliminate compromised cells.

MARKING SCHEME (3 Marks)
- Definition of both – 1.5 marks each

Q.9 What is the function of interleukin-2 in immune response?

Ans. Interleukin-2 (IL-2) is a cytokine secreted mainly by activated helper T cells, and it plays a pivotal role in immune regulation. IL-2 functions as a growth signal that stimulates the proliferation and differentiation of both T cells and B cells. It amplifies the immune response by promoting clonal expansion of cytotoxic T cells, which are necessary for killing infected or cancerous cells, and also enhances antibody production by B cells. Additionally, IL-2 helps sustain memory cell formation, ensuring long-term immunity. Because of its central role, IL-2 is sometimes used therapeutically in immunotherapy to boost immune function in certain diseases like cancer.

MARKING SCHEME (3 Marks)
- Definition – 1 mark; function – 2 marks

Q.10 Why are antipyretics used despite fever being a defence mechanism?

Ans. Fever is a natural defence mechanism that slows pathogen growth and boosts immune activity. However, when body temperature rises excessively, it can damage tissues, disrupt enzyme function, and place strain on vital organs such as the heart and brain. Antipyretics, such as paracetamol and ibuprofen, are used to safely lower dangerously high fevers. They act by inhibiting prostaglandin *synthesis* in the hypothalamus, which resets the elevated body temperature to normal. This prevents harmful effects of prolonged or extreme fever while allowing the immune system to continue fighting infection. Thus, antipyretics provide comfort and protection without negating the overall benefits of immune activity.

MARKING SCHEME (3 Marks)
- Definition – 1 mark; function – 2 marks

SLO [B-12-1-21]
Explain the role of memory cells in long term immunity.

ROLE OF MEMORY CELLS IN LONG-TERM IMMUNITY

One of the most remarkable features of the immune system is its ability to "remember" a pathogen once the body has been exposed to it. This capacity is due to the formation of **memory cells**, which belong to both B lymphocytes and T lymphocytes. Unlike short-lived effector cells that disappear once the infection is cleared, memory cells remain in the body for years or even a lifetime. Their presence ensures that the body is primed for a rapid and powerful response if the same pathogen tries to invade again.

1. Formation of Memory Cells

During the **primary immune response**, when the body first encounters a pathogen, B and T lymphocytes are activated. While most of these cells become short-lived effector cells to fight the infection, a portion differentiates into **long-lived memory cells**.

2. Persistence in the Body

Memory cells remain in the **lymphoid tissues and bloodstream** for years, sometimes for a lifetime. Unlike effector cells, they do not die quickly after the infection subsides, ensuring that the body retains "immunological memory."

3. **Rapid Secondary Immune Response**

If the same pathogen invades again, memory cells quickly recognize its antigens. They trigger a **secondary immune response**, which is:

- **Faster** (immediate recognition of antigen)
- **Stronger** (higher production of antibodies and cytotoxic T cells)
- **More specific** (better targeting of the same pathogen).

4. **Role in Vaccination and Long-Term Protection**

The effectiveness of **vaccines** relies on the formation of memory cells. By priming the immune system with harmless antigens, vaccines ensure that memory cells are available to mount a rapid defense if the real pathogen is encountered later.

Memory cells are the foundation of **active acquired immunity,** providing **long-term protection** against reinfection and ensuring that the immune system "remembers" past encounters.

SLO [B-12-1-20]
Draw the structural model of an antibody molecule.

STRUCTURAL MODEL OF AN ANTIBODY MOLECULE

Antibodies, also referred to as **immunoglobulins (Ig)**, are specialized **Y-shaped glycoproteins** produced by B lymphocytes in response to the presence of foreign antigens. They circulate in the bloodstream and lymphatic fluid and are transported to sites of infection where they bind specifically to antigens on pathogens, thereby facilitating their neutralization or destruction.

A typical antibody molecule consists of **four polypeptide chains:**

- Two identical heavy (H) chains
- Two identical light (L) chains

These chains are linked by **disulfide bonds**, giving rise to a characteristic Y-shaped structure. Each chain comprises three regions:

- **Variable (V) region:** Located at the amino-terminal ends of both heavy and light chains, the variable region contains unique amino acid sequences that determine antigen-binding specificity. Each antibody has **two antigen-binding sites,** one at the tip of each arm, allowing it to bind specifically to its target epitope.
- **Constant (C) region:** Found in the remaining portion of the polypeptide chains, the constant region has an amino acid sequence conserved within a specific immunoglobulin class (e.g., IgG, IgA). It mediates effector functions such as binding to immune cells or activation of the complement system.
- **Hinge region:** This flexible segment connects the arms (Fab regions) of the antibody to the stem (*Fc* region), allowing movement and spatial flexibility necessary for effective antigen binding.

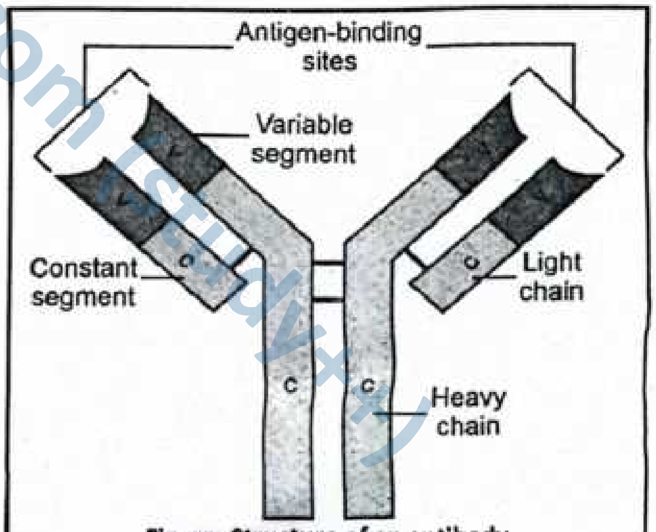

Figure: Structure of an antibody

The antibody can be divided functionally into:

- **Fab (Fragment antigen-binding) region:** Contains the variable regions that bind *specifically to antigens.*
- **Fc (Fragment crystallisable) region:** Comprises the constant *regions of the heavy chains* and interacts with immune effector cells and molecules, initiating immune responses.

Mode of Action of an Antibody

Antibodies contribute to immune defence through multiple mechanisms:

1. **Neutralization:** Antibodies bind directly to toxins or the surface proteins of viruses and bacteria, blocking their interaction with host cells and rendering them harmless.

2. **Opsonization:** Antibodies coat pathogens by forming antigen–antibody complexes, which enhances their recognition and uptake by phagocytes such as macrophages.
3. **Complement Activation:** The Fc region of bound antibodies can activate the **complement system,** resulting in the formation of the **membrane attack complex (MAC).** This leads to cell lysis or facilitates further phagocytosis.
4. **Antibody-Dependent Cell-Mediated Cytotoxicity (ADCC):** Antibodies bound to virus-infected or tumour cells recruit **natural killer (NK) cells,** which bind to the Fc region and induce apoptosis in the target cell.

SLO [B-12-1-22]
Define allergies and correlate the symptoms of allergies with the release of histamines.

ALLERGIES AND THE ROLE OF HISTAMINES

An **allergy** is an exaggerated immune response of the body to normally *harmless* substances, called **allergens**. Common allergens include pollen, dust, animal dander, insect stings, and certain foods. In allergic individuals, the immune system mistakenly identifies these allergens as threats and activates defense mechanisms that lead to uncomfortable or harmful symptoms.

Immunological Basis

When allergens enter the body, they stimulate the production of **IgE antibodies.** These antibodies bind to the surface of **mast cells** and **basophils**. Upon re-exposure to the same allergen, these cells rapidly release chemical mediators, most importantly **histamine,** into surrounding tissues and the bloodstream.

▸ Role of Histamine in Allergy Symptoms

Histamine is the principal chemical mediator released by mast cells and basophils during an allergic reaction. Once released, it binds to **histamine receptors (H1, H2, H3, H4)** on various tissues of the body, producing a wide range of physiological effects that account for most allergy symptoms.

1. **Vasodilation**: Histamine causes the smooth muscles of blood vessels to relax, leading to dilation of capillaries and small blood vessels. This results in increased blood flow to the affected area, which appears as **redness (erythema)** and **localized warmth**. This is especially noticeable in the skin during conditions like hives (urticaria).
2. **Increased Capillary Permeability**: By making the endothelial lining of capillaries more permeable, histamine allows plasma proteins and fluid to escape into surrounding tissues. This produces **swelling (edema)** and a feeling of puffiness in allergic sites, such as swollen eyelids or lips after allergen exposure.
3. **Smooth Muscle Contraction**: In the respiratory tract, histamine stimulates contraction of smooth muscles in the bronchioles. This narrows the airways and results in **difficulty breathing, wheezing, and chest tightness**, commonly seen in allergic asthma. In the gastrointestinal tract, smooth muscle contraction can cause abdominal cramps and diarrhea in food allergies.
4. **Stimulation of Nerve Endings**: Histamine directly irritates sensory nerve endings, giving rise to sensations of **itching (pruritus)** and irritation. This explains the intense itch associated with skin allergies such as eczema and insect bite reactions.
5. **Excess Mucus Secretion**: Histamine promotes secretion of *mucus* from epithelial cells lining the respiratory tract. This leads to **nasal congestion, runny nose (rhinorrhea), sneezing, and watery eyes,** all characteristic features of hay fever (allergic rhinitis).

▸ *Correlation* of Symptoms with Histamine Release

The release of histamine during an allergic reaction explains the wide range of symptoms observed:

- **Sneezing, watery eyes, and runny nose**: These symptoms occur due to histamine-induced irritation of mucous membranes in the nose and eyes. At the same time, histamine increases mucus secretion, which contributes to nasal discharge and eye watering.
- **Swelling and redness of skin (hives/urticaria)**: Histamine causes vasodilation of small blood vessels and increases capillary permeability, allowing fluid to leak into the tissues. This results in localized swelling, raised patches (wheals), and redness of the skin.

- **Difficulty in breathing (allergic asthma):** In the respiratory tract, histamine triggers contraction of smooth muscles in the bronchioles. This airway narrowing makes breathing difficult and leads to wheezing, chest tightness, and shortness of breath.

Thus, allergies occur due to the body's **hypersensitive reaction** to harmless substances. The release of **histamines** from mast cells directly explains the symptoms such as redness, itching, swelling, and respiratory distress.

SLO [B-12-1-23]
Describe the autoimmune diseases with examples.

AUTOIMMUNE DISEASE

Autoimmune diseases are disorders in which the body's immune system mistakenly attacks its own healthy cells, tissues, or organs. Normally, the immune system recognizes "self" (body's own cells) and "non-self" (foreign invaders like bacteria or viruses). In autoimmunity, this recognition fails, and the immune system mounts a destructive response against self-antigens.

Mechanism

The development of autoimmune diseases is primarily due to a failure in the immune system's ability to recognize the difference between self and non-self. Normally, immune cells undergo a selection process to eliminate or inactivate those that react against the body's own tissues. When this system of regulation breaks down, the following events occur:

- **Loss of Self-Tolerance:** The immune system loses its ability to tolerate self-antigens. As a result, body tissues are mistakenly identified as foreign.
- **Immune Response Against Self:** Both lymphocytes (especially T cells) and autoantibodies mount an immune attack against the body's own cells. This leads to persistent inflammation, destruction of tissues, and impaired organ function.
- **Chronic Nature of the Disorder:** Since the self-antigens remain in the body, the immune response becomes continuous and recurring. This explains why autoimmune diseases are usually long-lasting, progressive, and difficult to cure completely.

Examples of Autoimmune Diseases

1. **Rheumatoid Arthritis (RA):** In this chronic autoimmune disorder, the immune system mistakenly attacks the **synovial membranes** lining the joints. This causes persistent **inflammation, pain, stiffness, and swelling**. Over time, cartilage and bone erosion can occur, leading to **joint deformity and loss of mobility.** RA may also produce systemic effects, such as fatigue and cardiovascular complications.
2. **Type 1 Diabetes Mellitus:** Here, **cytotoxic T cells** target and destroy the **β-cells of the pancreas**, which are responsible for insulin secretion. The destruction of these cells leads to ***insulin deficiency***, causing elevated blood glucose levels (**hyperglycemia**). If untreated, this condition results in severe metabolic disturbances, and patients require lifelong insulin therapy.
3. **Multiple Sclerosis (MS):** In MS, the immune system attacks the **myelin sheath** that insulates nerve fibers in the brain and spinal cord. Loss of myelin disrupts nerve conduction, leading to **muscle weakness, numbness, impaired coordination, and vision problems**. In advanced cases, paralysis and cognitive decline may develop.
4. **Systemic Lupus Erythematosus (SLE):** SLE is a **multisystem autoimmune disease** in which autoantibodies attack tissues and organs throughout the body, including the **skin, kidneys, joints, blood vessels, and heart.** Its hallmark symptom is a **"butterfly-shaped" rash** across the cheeks and nose. Other manifestations include **joint pain, anemia, kidney dysfunction, and extreme fatigue.**
5. **Vitiligo:** Vitiligo is an autoimmune skin disorder in which the immune system destroys melanocytes, the pigment-producing cells in the skin. As a result, patients develop **patches of depigmented (white) skin**, often symmetrically distributed. While vitiligo does not cause physical pain, it can have significant **psychological and social impact** due to visible skin changes.

DO YOU KNOW?
The first antibody-based therapies were used to treat snake bites and were called *antivenoms!*

Autoimmune diseases arise from a breakdown in the body's ability to recognize its own tissues as "self." This leads to self-destruction by immune cells and antibodies, causing chronic and often debilitating conditions. Early diagnosis and management are important to control immune overactivity and prevent long-term damage.

Figure: Vitiligo causes pigment-less patches on skin

SLOs Based (MCQs)

1. **What is the primary reason for administering immunosuppressive drugs to transplant recipients?**
A. To improve blood circulation
B. To reduce allergic responses
C. To prevent immune rejection of donor tissue
D. To promote cell regeneration

2. **Transplant rejection is primarily mediated by:**
A. Antibodies only B. T lymphocytes
C. Platelets D. Natural killer cells

3. **Which of the following best describes transplant rejection?**
A. Autoimmune response against self-antigens
B. Non-specific inflammatory response
C. Delayed hypersensitivity reaction mediated by lymphocytes
D. Immediate allergic reaction

4. **Which cells are primarily responsible for recognizing and attacking foreign transplanted tissues?**
A. Basophils B. Erythrocytes
C. T helper cells D. Neutrophils

5. **Which type of immunity is suppressed by immunosuppressive drugs in transplant patients?**
A. Passive immunity B. Innate immunity
C. Adaptive immunity D. Artificial immunity

6. **Immunosuppressive drugs enhance:**
A. Graft tolerance B. Immune sensitivity
C. Histamine release D. Natural killer cell activity

7. **Which of the following is a common side effect of immunosuppressive therapy?**
A. Hyperglycemia B. Hypotension
C. Leukopenia D. Hyperthyroidism

8. **Which condition may result from immune-suppressant-induced suppression of immune surveillance?**
A. Stroke B. Lymphoma
C. Scurvy D. Osteoporosis

9. **Which electrolyte imbalance is commonly associated with immunosuppressive therapy?**
A. Hypokalemia B. Hypernatremia
C. Hyperkalemia D. Hypocalcemia

10. **One major long-term risk of immunosuppressive therapy in transplant recipients is:**
A. Dehydration B. Autoimmunity
C. Opportunistic infections D. Hormonal imbalance

11. **Which of the following symptoms is directly linked to neurotoxicity caused by immuno-suppressants?**
A. Coughing B. Tremors
C. Skin blisters D. Weight loss

12. **Why is lifelong monitoring necessary for organ transplant recipients?**
A. To ensure drug tolerance
B. To prevent pain recurrence
C. To detect rejection or drug side effects early
D. To avoid blood pressure changes

13. **Which immune molecules are primarily involved in antibody-mediated rejection?**
A. Cytokines B. Histamines
C. Antibodies D. Interferons

14. **Which class of drug is commonly used in transplant recipients to inhibit T-cell activation?**
A. Beta blockers B. Corticosteroids
C. Calcineurin inhibitors D. Antibiotics

15. Graft rejection due to antibody involvement is called:

A. Humoral rejection B. Autoimmunity
C. Innate reaction D. Active immunity

16. What component of the donor tissue is primarily recognized as foreign by the host immune system?

A. Hormones B. DNA
C. MHC antigens D. Enzymes

17. What is the role of lymphokines during transplant rejection?

A. Neutralize toxins
B. Promote antibody production
C. Recruit and activate immune cells
D. Reduce inflammation

18. Which of the following is not a known side effect of immunosuppressive drugs?

A. Hypertension B. Hyperkalemia
C. Seizures D. Hypoglycemia

19. The immune suppression caused by these drugs often leads to:

A. Improved metabolism
B. Decreased allergic response
C. Increased susceptibility to infections
D. Better oxygen uptake

20. What organ transplant is most commonly associated with the use of immunosuppressants?

A. Brain B. Heart
C. Skin D. Pancreas

21. Which condition is not caused directly by immunosuppressant side effects?

A. Leukopenia B. Thrombocytopenia
C. Astigmatism D. Neurotoxicity

22. Which immune cell helps B cells produce antibodies during rejection?

A. Macrophage B. Helper T cell
C. Plasma cell D. Natural killer cell

23. Pain, fatigue, and weakness in transplant recipients are often due to:

A. Infection only B. Surgical trauma
C. System-wide drug effects D. Vitamin deficiency

24. Histocompatibility refers to:

A. Compatibility of hormones
B. Compatibility of skin tone
C. Compatibility of MHC antigens
D. Compatibility of blood volume

25. Anaphylactic reactions in transplant patients are most likely caused by:

A. Nervous system injury B. High fever
C. Drug hypersensitivity D. Organ swelling

26. Why are antibodies insufficient in preventing transplant rejection alone?

A. They are not specific to MHC
B. T cells are primarily responsible
C. Antibodies are not produced post-transplant
D. Antibodies enhance tissue growth

27. Which organ system is commonly affected by neurotoxic side effects of immunosuppressants?

A. Respiratory system B. Nervous system
C. Muscular system D. Digestive system

28. What is the term for drugs that reduce immune function to allow graft acceptance?

A. Antibiotics B. Immunostimulants
C. Immunosuppressants D. Cytotoxins

29. A transplant recipient who develops tremors and hallucinations is likely experiencing:

A. Acute rejection B. Bacterial infection
C. Neurotoxicity D. Autoimmune flare-up

30. Which of the following statements best describes immunosuppressive therapy?

A. It boosts innate immunity to resist infections.
B. It suppresses the immune system to prevent graft rejection.
C. It prevents the production of digestive enzymes.
D. It strengthens the blood-brain barrier.

Answer Key

1. C	2. B	3. C	4. C	5. C	6. A	7. C	8. B	9. C	10. C	11. B	12. C	13. C	14. C	15. A
16. C	17. C	18. D	19. C	20. B	21. C	22. B	23. C	24. C	25. C	26. B	27. B	28. C	29. C	30. B

Short Questions

Q.1 Explain why transplant recipients are given immunosuppressive drugs.

Ans. Immunosuppressive drugs are administered after organ transplantation to prevent the recipient's immune system from recognizing donor tissue as non-self and mounting an attack that would destroy the graft. The principal target is T-cell mediated immunity: drugs (e.g., calcineurin inhibitors, corticosteroids, antiproliferatives) reduce antigen presentation, inhibit T-cell activation/proliferation and blunt cytokine production. By lowering cellular and humoral immune responses, these agents curb acute and chronic rejection, improving graft survival and function. However, suppression must be balanced to avoid under-immunosuppression (rejection) or over-immunosuppression (infection, malignancy). Therapeutic regimens are therefore individualized, often combining agents with complementary mechanisms to maximize graft tolerance while minimizing adverse effects.

> **MARKING SCHEME (3 Marks)**
> - Reason for immunosuppression (prevent immune recognition / rejection) — 2.00 marks; principal effects (reduce T-cell activity / promote graft survival) — 1.00 mark.

Q.2 Define transplant rejection and identify the immune cells primarily involved.

Ans. Transplant rejection is the immune-mediated destruction or dysfunction of transplanted tissue because the recipient's immune system recognizes donor antigens (mainly mismatched MHC molecules) as foreign. Rejection can be classified clinically as hyperacute (antibody-mediated immediately), acute (cell-mediated within weeks–months), or chronic (progressive vascular and fibrotic damage). The key immune effectors are T lymphocytes—helper ($CD4^+$) T cells that orchestrate responses and cytotoxic ($CD8^+$) T cells that directly kill graft cells. B cells and plasma cells generate alloantibodies contributing to *antibody-mediated rejection*, while macrophages and NK cells also participate in tissue injury. Accurate identification of the pathways guides immunosuppressive strategy and graft management.

> **MARKING SCHEME (3 Marks)**
> - Definition of transplant rejection — 2.00 marks; names/identification of immune cells involved — 1.00 mark.

Q.3 Explain how immunosuppressive drugs increase the risk of infection; give examples of opportunistic infections.

Ans. Immunosuppressive drugs blunt innate and adaptive immune defenses essential for controlling pathogens. By inhibiting T-cell activation, reducing cytokine signalling, impairing neutrophil function, or decreasing antibody responses, these agents lower the body's ability to detect and clear microbes. As a result, patients become susceptible to opportunistic organisms that are typically controlled by intact immunity. Common opportunistic infections in transplant recipients include **cytomegalovirus (CMV)** and **Epstein–Barr virus (EBV)** (viral), **Pneumocystis jirovecii** (fungal/atypical), invasive **Candida** or **Aspergillus** species (fungal), and bacterial sepsis or **Listeria**. Prophylactic antimicrobials and close monitoring are therefore standard parts of post-transplant care.

> **MARKING SCHEME (3 Marks)**
> - Reason (how immunosuppression predisposes to infection) — 1 mark; names/examples of opportunistic infections (any two or more) — 2 marks.

Q.4 List common side effects of immunosuppressive therapy and explain their clinical relevance.

Ans. Immunosuppressive regimens produce multiple adverse effects because they affect normal physiological systems. Common problems include **leukopenia** (reduced white cells → increased infection risk), **nephrotoxicity** (particularly with calcineurin inhibitors → impaired renal function), **hypertension** and **hyperkalemia** (electrolyte and cardiovascular effects), **gastrointestinal disturbances** such as diarrhoea, **neurotoxicity** (tremor, seizures), metabolic derangements (hyperglycaemia), and a higher long-term risk of **malignancy** (e.g., post-transplant lymphoproliferative disorder). Clinically, these side effects require dose adjustments, switching agents, and proactive monitoring (blood counts, renal function, blood pressure, malignancy screening) to maintain patient safety while preserving graft function.

> **MARKING SCHEME (3 Marks)**
> - Any three common side effects (1.00 mark each; e.g., leukopenia, nephrotoxicity, increased cancer risk).

Q.5 Describe, step by step, how helper T cells contribute to transplant rejection.

Ans. Helper T ($CD4^+$) cells drive rejection through a sequence of events. (1) ***Recognition:*** donor antigens (directly on graft antigen-presenting cells or indirectly after host APCs present processed donor peptides) are recognized by $CD4^+$ T-cell receptors. (2) **Activation:** costimulatory signals and antigen recognition **activate** helper T cells, triggering proliferation (clonal expansion). (3) **Cytokine secretion: activated helper T cells** release cytokines (e.g., IL-2, IFN-γ) that amplify immune responses. (4) **Effector recruitment:** cytokines activate cytotoxic $CD8^+$ T cells, macrophages, and B cells; B cells differentiate into plasma cells producing alloantibodies. (5) **Tissue injury:** cytotoxic cells, antibody-dependent mechanisms, and inflammatory mediators attack graft cells, causing cell death, inflammation and vascular damage leading to graft dysfunction.

> **MARKING SCHEME (3 Marks)**
> - Clear stepwise mechanism covering recognition → activation → cytokine release → effector activation → graft injury — total 3.00 marks.

Q.6 Explain the role of B cells in transplant rejection.

Ans. B cells play a central role in antibody-mediated (humoral) transplant rejection. Once helper T cells recognize donor MHC antigens, they activate B cells through cytokines and costimulatory signals. Activated B cells differentiate into plasma cells, which secrete alloantibodies directed against donor antigens. These antibodies initiate several damaging processes: (1) **complement activation**, leading to cell lysis and inflammation; (2) **opsonization**, which enhances phagocytosis of graft cells by macrophages and neutrophils; and (3) **vascular injury**, as antibodies target endothelial cells, causing thrombosis and ischemia. Collectively, these processes compromise blood supply and tissue integrity, resulting in graft dysfunction and failure if uncontrolled.

> **MARKING SCHEME (3 Marks)**
> - Stepwise mechanism — activation by helper T cells → antibody production → complement activation / phagocytosis / vascular injury.

Q.7 Explain why long-term immunosuppressive therapy often leads to fatigue and weakness.

Ans. Chronic use of immunosuppressive drugs affects multiple organ systems, producing cumulative effects that manifest as fatigue and weakness. **Hematological effects** such as leukopenia or anemia reduce oxygen transport and immune capacity. **Gastrointestinal effects** like diarrhoea and poor nutrient absorption impair energy levels. **Neurological effects** including tremors, headaches, or neurotoxicity directly contribute to exhaustion. Additionally, suppressed immunity predisposes patients to recurrent or chronic infections, which keep the body in a state of physiological stress. Persistent inflammation, metabolic derangements (e.g., electrolyte imbalance, hyperglycaemia), and drug-related toxicity further drain energy reserves. Together, these interconnected disturbances explain the chronic fatigue seen in transplant patients on long-term immunosuppressive regimens.

> **MARKING SCHEME (3 Marks)**
> - Names of affected systems — 2.00 marks; other harmful contributing effects — 1.00 mark.

Q.8 Describe how modern medicine has improved transplant outcomes.

Ans. Modern medicine has transformed transplantation into a highly successful clinical practice. First, **advances in surgical techniques** ensure safer operations with better vascular connections, reducing complications during and after surgery. Second, **precise tissue typing and cross-matching** of donor and recipient MHC antigens minimize immunological incompatibility, lowering the risk of rejection. Third, the **development of targeted immunosuppressive drugs** (e.g., calcineurin inhibitors, corticosteroids, antiproliferatives, biologics) has significantly improved graft survival by controlling rejection episodes. Additional innovations such as improved organ preservation methods, post-transplant monitoring, and infection prophylaxis have further boosted success rates. Collectively, these improvements have made organ transplantation routine, with many patients achieving long-term survival and improved quality of life.

> **MARKING SCHEME (3 Marks)**
> - Names of three main improvements — surgical methods, tissue typing, immune-suppressive drugs (1 mark each).

Q.9 What precautions are necessary during immunosuppressive therapy, and why are they important?

Ans. Patients on immunosuppressive therapy require continuous medical supervision to balance graft protection against harmful side effects. Regular monitoring includes **infection surveillance** (blood tests, cultures, imaging) because lowered immunity increases vulnerability to opportunistic pathogens. **Blood pressure and electrolyte checks** are essential since drugs like calcineurin inhibitors can cause hypertension and hyperkalemia. **Renal and hepatic function tests** detect early toxicity, preventing organ damage. Adjusting drug dosage and adding prophylactic antibiotics, antivirals, or antifungals are common strategies. Vaccination (non-live) and lifestyle precautions, such as hygiene and avoiding exposure to infections, also support safety. These precautions help maximize graft survival while reducing complications, enabling patients to maintain long-term health and quality of life.

> **MARKING SCHEME (3 Marks)**
> - Relevant precautions with explanation of their effects/advantages — 3.00 marks.

Q.10 Explain how the immune system recognizes transplanted tissue as foreign.

Ans. The immune system identifies transplanted tissue as foreign primarily through differences in **major histocompatibility complex (MHC) molecules**. Donor cells express MHC antigens that are distinct from those of the recipient. Host **antigen-presenting cells** or donor-derived cells present these foreign antigens to recipient helper T cells. This recognition activates helper T cells, which release cytokines that

> **MARKING SCHEME (3 Marks)**
> - Mechanism of recognition via foreign MHC → T-cell activation → immune cascade.

stimulate cytotoxic T cells, B cells, and macrophages. Cytotoxic T cells attack graft cells directly, while B cells produce antibodies that initiate antibody-mediated rejection. Macrophages and other immune effectors amplify tissue damage through inflammation. This cascade of cellular and humoral responses ultimately leads the immune system to perceive the graft as a threat, triggering rejection.

Exercise

I Multiple Choice Questions (MCQs)

❖ Select the correct Answer.

1. **Plasma cells are**
   A. the same as memory cells
   B. formed from blood plasma
   C. B cells that are actively secreting antibody
   D. inactive T cells carried in the plasma
2. **Antibodies combine with antigens**
   A. at variable regions
   B. at constant region
   C. only if macrophages are present
   D. both A and C are correct
3. **In addition to the immune system, we are protected from disease by**
   A. normal body temperature
   B. hormones
   C. antigens
   D. mucous membrane and cilia
4. **Fever decrease the**
   A. interferon production
   B. concentration of iron in the blood
   C. activity of phagocytes
   D. inflammation
5. **T and B cells are**
   A. lymphocytes B. macrophages
   C. natural killer cells D. red blood cells
6. **When B-cells are presented with antigen they differentiate into**
   A. T-cells B. helper T-cells
   C. plasma cells D. bursa cells
7. **When one receives a booster shot for polio which type of cell is most directly stimulated?**
   A. killer T-cells B. memory cells
   C. phagocytes D. suppressor cells
8. **Natural killer cells kill virus infected and cancerous cells by:**
   A. phagocytosis B. complement proteins
   C. perforins and granzymes D. antibodies
9. **Colostrum provides**
   A. natural active immunity
   B. artificial active immunity
   C. natural passive immunity
   D. artificial passive immunity
10. **mAbs can help to combat diseases like:**
    A. measles B. cancer
    C. anorexia D. haemophilia
11. **A patient has a severe burn injury, compromising their skin barrier. Which of the following is most likely to occur?**
    A. Increased risk of infection due to loss of chemical and physical barriers
    B. Decreased risk of infection due to increased immune response
    C. No change in infection risk
    D. Increased risk of autoimmune disease
12. **Dendritic cells are crucial for initiating an adaptive immune response. What would happen if dendritic cells were unable to present antigens?**
    A. The adaptive immune response would be enhanced
    B. The innate immune response would be impaired
    C. The adaptive immune response would be impaired
    D. There would be no impact on the immune response
13. **Neutrophils use extracellular traps to kill pathogens. What is the primary advantage of this mechanism?**
    A. Targeted killing of specific pathogens
    B. Ability to phagocytose large pathogens
    C. Immobilization of pathogens in a localized area
    D. Activation of complement system
14. **NK cells play a crucial role in controlling viral infections and cancer. What would happen if NK cell function was impaired?**
    A. Increased risk of bacterial infections
    B. Increased risk of viral infections and cancer
    C. Decreased risk of autoimmune disease
    D. No impact on immune function

15. **Interferons released by virus-infected cells trigger a response in neighboring cells. What is the primary purpose of this response?**

A. To induce apoptosis in uninfected cells
B. To destroy RNA in infected cells
C. To activate the adaptive immune response
D. To prevent viral replication in neighboring cells

16. **Antibodies can bind to an antigen, forming an antigen-antibody complex. What would be the most likely outcome if antibodies were unable to opsonize bacteria?**

A. Increased phagocytosis of bacteria
B. Decreased phagocytosis of bacteria
C. Increased activation of complement system
D. Decreased activation of NK cells

17. **Monoclonal antibodies have revolutionized the field of immunotherapy. What is a potential benefit of using monoclonal antibodies in cancer treatment?**

A. Increased killing of virus infected cells
B. Targeted killing of cancer cells
C. Enhanced tumor growth
D. Increased risk of infection

18. **Vitiligo is an autoimmune disease characterized by skin depigmentation. What would be a potential consequence of vitiligo, in majority of patients, if left untreated?**

A. Increased risk of skin cancer
B. Decreased risk of autoimmune disease
C. Increased risk of infection
D. Emotional distress due to cosmetic impact

19. **Transplant rejection occurs due to the immune system's ability to recognize foreign antigens. What would happen if a patient received a transplant from an identical twin?**

A. Increased risk of graft-versus-host disease
B. Decreased risk of transplant rejection
C. Increased risk of autoimmune disease
D. No impact on immune function

20. **Lymphokines, such as interleukins, play a crucial role in coordinating the immune response. What would be the consequence of impaired lymphokine production?**

A. Enhanced activation of cytotoxic T-cells
B. Reduced activation of immune cells
C. Increased production of antibodies
D. Enhanced phagocytic activity of neutrophils

Answer Key

1. C	2. A	3. D	4. B	5. A	6. C	7. B	8. C	9. C	10. B
11. A	12. C	13. C	14. B	15. D	16. B	17. B	18. D	19. B	20. B

II Short Answer Questions

Q.1 What is immune response?

Ans. The immune response is the coordinated defensive reaction of the body against harmful agents such as bacteria, viruses, fungi, and toxins. It begins when foreign antigens are recognized by immune receptors. This triggers activation of innate immunity, including phagocytes, natural killer cells, and inflammation, which act rapidly but non-specifically. If the threat persists, the adaptive immune response is activated. B lymphocytes produce antibodies that bind and neutralize pathogens, while T lymphocytes either assist other immune cells (helper T cells) or directly destroy infected cells (cytotoxic T cells). The adaptive system also develops *immunological memory*, ensuring faster and stronger responses upon re-exposure. Thus, immune responses maintain health by eliminating invaders and preventing disease progression.

> **MARKING SCHEME (3 Marks)**
> - Definition of immune response — 1.5 marks; description of mechanisms and significance — 1.5 marks.

Q.2 Name the three lines of defence against microbial attack.

Ans. *The human* body protects itself against infections through three levels of defence. The **first line of defence** consists of physical and chemical barriers, such as intact skin, mucous membranes, saliva, tears, stomach acid, and *secretions from sweat and oil glands*, which prevent pathogen *entry*. The **second line of defence** is the innate immune response, a non-specific mechanism including phagocytes, natural killer cells, inflammation, and fever. These act quickly once pathogens bypass external barriers. The **third line of defence** is the adaptive immune response,

> **MARKING SCHEME (3 Marks)**
> - Correct identification of first line — 1 mark; second line — 1 mark; third line — 1 mark.

mediated by B and T lymphocytes. This system is highly specific, capable of antigen recognition, antibody production, and memory formation. Together, these defences form a layered protection that ensures both immediate and long-term immunity.

Q.3 How oil and sweat glands take part in defence against microorganisms?

Ans. Oil and sweat glands are important components of the body's first line of defence against microbial invasion. **Sebaceous (oil) glands** secrete sebum onto the skin surface. Sebum contains fatty acids that lower the pH of the skin, creating an acidic environment that inhibits the growth of most bacteria and fungi. This biochemical barrier is particularly effective in preventing colonization of harmful microbes. **Sweat glands** produce sweat, which contains antimicrobial substances such as lysozyme. Lysozyme is an enzyme capable of breaking down the peptidoglycan layer of bacterial cell walls, leading to bacterial lysis. Additionally, sweat helps wash away microbes mechanically. Together, oil and sweat secretions form a protective chemical barrier that supports skin's role as a physical shield.

> **MARKING SCHEME (3 Marks)**
> - Role of sebum — 1 mark; role of sweat/lysozyme — 1 mark; combined protective effect — 1 mark.

Q.4 Name the parts of antibody molecule.

Ans. An antibody molecule has a characteristic Y-shaped structure composed of **two heavy chains** and **two light chains**, held together by disulfide bonds. Each antibody has two distinct regions. The **variable regions** are located at the tips of the Y-shape. They differ in amino acid sequence among antibodies and form the antigen-binding sites, allowing recognition of specific antigens with high precision. The **constant regions**, found in the stem and lower parts of the Y, are similar among antibodies of the same class and determine their effector function, such as complement activation or binding to immune cells. This structure enables antibodies to both specifically recognize pathogens and recruit other immune mechanisms for pathogen destruction.

> **MARKING SCHEME (3 Marks)**
> - Variable regions/antigen-binding site — 1 mark; constant region and its role — 1 mark; heavy and light chain structure — 1 mark.

Q.5 How interferons control spread of virus in the body?

Ans. Interferons are specialized proteins produced by virus-infected cells to protect neighboring cells from viral spread. When a cell is invaded by a virus, it releases interferons into the surrounding tissue. These molecules bind to receptors on adjacent uninfected cells, stimulating them to synthesize antiviral proteins. These proteins interfere with viral replication by degrading viral RNA or inhibiting protein synthesis. In addition to direct antiviral activity, interferons enhance the immune response by stimulating natural killer (NK) cells and macrophages, which target and destroy infected cells. They also increase antigen presentation, making infected cells more visible to cytotoxic T cells. Thus, interferons act as important messengers in the innate immune system, containing viral infections until adaptive immunity develops.

> **MARKING SCHEME (3 Marks)**
> - Definition of interferons — 1 mark; mechanism of antiviral action — 1 mark; immune-enhancing role — 1 mark.

Q.6 How do antibodies kill pathogens?

Ans. Antibodies defend the body by binding specifically to antigens present on the surface of pathogens, forming antigen–antibody complexes. This binding can neutralize pathogens directly by blocking their attachment to host cells, rendering them harmless. Antibodies also facilitate **opsonization**, where pathogens are "tagged" for easier recognition and engulfment by phagocytes such as macrophages and neutrophils. Another major mechanism is activation of the **complement system**, a cascade of plasma proteins that results in membrane damage and lysis of invading microbes. Additionally, antibodies can cause **agglutination**, where multiple pathogens are clumped together, making them more easily removed by phagocytosis. Through these combined actions, antibodies play a central role in adaptive immunity.

> **MARKING SCHEME (3 Marks)**
> - Neutralization — 1 mark; opsonization and complement activation — 1 mark; agglutination/combined effect — 1 mark.

Q.7 Why does each antibody bind only to a specific antigen?

Ans. Each antibody molecule contains a variable region located at the antigen-binding site, which is uniquely shaped due to differences in amino acid sequences. This region is complementary to a specific **epitope**, the precise portion of an antigen that it binds. The lock-and-key type interaction ensures **specificity**, allowing an antibody to

> **MARKING SCHEME (3 Marks)**
> - Structure of variable region — 1 mark; binding to epitope/specificity — 1 mark; importance of specificity — 1 mark.

recognize and attach only to its corresponding antigen. This prevents cross-reactions with the body's own cells and ensures precise targeting of pathogens. Specific binding also initiates appropriate immune responses such as neutralization, opsonization, or complement activation. Without this high degree of specificity, immune responses would be random and harmful. Thus, antibody–antigen specificity is essential for effective and safe immunity.

Q.8 Why is passive immunity temporary?

Ans. Passive immunity arises when pre-formed antibodies are transferred into the body from an external source, such as maternal antibodies crossing the placenta, antibodies in breast milk, or therapeutic antibody injections. Unlike active immunity, the recipient's immune system does not produce its own antibodies or memory cells. The borrowed antibodies provide **immediate but short-lived protection** against infection. Over time, these externally supplied antibodies are naturally broken down and eliminated by the body. Since no immunological memory is established, the individual remains unprotected once antibody levels decline. Therefore, passive immunity is temporary, serving as a short-term defense mechanism until the individual's own adaptive immune system develops or an active vaccination can be administered.

> **MARKING SCHEME (3 Marks)**
> - Definition of passive immunity — 1 mark; absence of memory cells — 1 mark; reason for temporary protection — 1 mark.

Q.9 Name the disorders of immune system.

Ans. The immune system may malfunction, resulting in several disorders. One group is **autoimmune diseases**, where the immune system attacks self-tissues. Examples include systemic lupus erythematosus, rheumatoid arthritis, type 1 diabetes, and multiple sclerosis. Another category is **immunodeficiency disorders**, in which the immune system fails to respond adequately. These may be primary (genetic) or secondary (acquired), such as AIDS caused by HIV infection. **Allergies and hypersensitivity reactions** represent exaggerated immune responses to harmless substances like pollen, dust, or food. Finally, complications such as **graft rejection** occur when the immune system recognizes transplanted tissue as foreign. Each of these conditions reflects an imbalance in immune regulation, either overactivity, underactivity, or misdirected activity.

> **MARKING SCHEME (3 Marks)**
> - Autoimmune diseases — 1 mark; immunodeficiency disorders — 1 mark; allergies/transplant rejection — 1 mark.

Q.10 What are the autoimmune diseases?

Ans. Autoimmune diseases are medical conditions in which the immune system loses **self-tolerance** and begins to recognize the body's own cells and tissues as foreign. As a result, it launches immune attacks against healthy organs, leading to chronic inflammation and tissue damage. The exact causes may involve genetic predisposition, molecular mimicry, or environmental triggers that confuse immune recognition. For example, in **type 1 diabetes**, immune cells destroy insulin-producing pancreatic cells. In **multiple sclerosis**, myelin sheaths of nerve cells are damaged. Other examples include **rheumatoid arthritis** (joints), **vitiligo** (skin), and **systemic lupus erythematosus** (multi-organ). Since self-antigens are constantly present, autoimmune diseases are usually long-lasting and require medical management to suppress overactive immunity.

> **MARKING SCHEME (3 Marks)**
> - Definition/self-tolerance failure — 1 mark; mechanism/immune attack on tissues — 1 mark; examples — 1 mark.

Q.11 Write the differences between:

(a) Sebaceous Glands and Sweat Glands

Ans. Sebaceous glands and sweat glands are both exocrine glands of the skin but differ in structure and function. **Sebaceous glands** secrete an oily substance called sebum, which lubricates skin and hair, and lowers skin pH to inhibit microbial growth. They usually open into hair follicles and are distributed throughout the body, except on the palms and soles.

In contrast, **sweat glands** produce sweat, composed mainly of water, salts, and small amounts of urea. Sweat is released directly onto the skin surface and plays a major role in thermoregulation through evaporative cooling. It also aids in excretion and contains lysozyme, providing antimicrobial defense. Thus, sebaceous glands protect and condition, while sweat glands regulate temperature and excrete wastes.

> **MARKING SCHEME (3 Marks)**
> - Secretion and function — 2 mark; opening/distribution — 1 mark; role in immunity — 1 mark.

(b) Macrophages and Neutrophils

Ans. Macrophages and neutrophils are both phagocytic white blood cells but differ in origin, lifespan, and role. **Neutrophils** arise from myeloid stem cells in bone marrow and are short-lived, surviving only a few days. They circulate in the blood and are the first responders during infection, rapidly engulfing and destroying pathogens.

> **MARKING SCHEME (3 Marks)**
> - Origin/lifespan — 1 mark; site/presence — 1 mark; role in immunity — 1 mark.

Macrophages, on the other hand, are long-lived cells derived from circulating monocytes that migrate into tissues. Besides engulfing pathogens, they also process and present antigens to T lymphocytes, linking innate and adaptive immunity. Macrophages remain resident in tissues like the liver, lungs, and spleen, providing long-term defense, whereas neutrophils specialize in acute, rapid responses at infection sites. Together, they ensure both immediate and sustained immunity.

(c) Antibody-mediated and Cell-mediated Immune Responses

Ans. The immune system operates through two complementary pathways. The **antibody-mediated immune response** involves B lymphocytes, which, upon activation, differentiate into plasma cells that secrete antibodies. These antibodies neutralize extracellular pathogens like bacteria, viruses in body fluids, and toxins, and mark them for destruction by phagocytes or complement proteins. Memory B cells also form, ensuring faster responses on re-exposure.

> **MARKING SCHEME (3 Marks)**
> - Type of lymphocytes — 1 mark; target/effector mechanism — 1 mark; memory cells/role — 1 mark.

In contrast, the **cell-mediated immune response** is carried out by T lymphocytes, mainly cytotoxic T cells. This pathway targets **intracellular pathogens**, such as viruses within host cells, and abnormal cells like cancerous ones. Cytotoxic T cells directly kill infected cells by releasing perforin and granzymes. Memory T cells provide long-term surveillance, ensuring lasting immunity against intracellular threats.

(d) Cytotoxic T Cells and Suppressor T Cells

Ans. Cytotoxic T cells (Tc) and suppressor T cells (Ts), also called regulatory T cells, perform opposite roles in immunity. **Cytotoxic T cells** are $CD8^+$ cells activated by antigen presentation with MHC class I molecules. Once activated, they directly kill virus-infected cells and tumor cells by releasing perforin and granzymes, which induce lysis and apoptosis. They are critical in defending against intracellular pathogens and cancers.

> **MARKING SCHEME (3 Marks)**
> - Function — 1 mark; markers/action — 1 mark; role/activation — 1 mark.

In contrast, **suppressor T cells** ($CD4^+$ or $CD25^+$ phenotype) function primarily to regulate the immune system. They inhibit overactivation of immune cells, preventing excessive inflammation and autoimmune reactions. Ts cells become active in later stages of an immune response, ensuring balance between defense and tolerance. Thus, Tc destroy threats, while Ts prevent self-damage.

(e) Plasma Cells and Memory Cells

Ans. Plasma cells and memory cells are both derived from activated B lymphocytes but serve distinct functions. **Plasma cells** are short-lived effector cells that immediately produce and secrete large quantities of antibodies to combat current infections. They play a critical role during the first exposure to a pathogen, but once the infection resolves, most plasma cells die off.

> **MARKING SCHEME (3 Marks)**
> - Origin — 1 mark; function/lifespan — 1 mark; role in immunity — 1 mark.

In contrast, **memory cells** (both B and T types) are long-lived and persist in the body after the infection subsides. They do not actively secrete antibodies but retain information about the specific antigen. Upon re-exposure, memory cells rapidly differentiate and trigger a stronger, faster immune response, ensuring long-term immunity. Plasma cells provide immediate defense, while memory cells ensure future protection.

(f) Antibodies and Antigens

Ans. Antibodies and antigens are complementary components of the immune system but differ fundamentally in nature and function. **Antibodies** are specialized Y-shaped proteins produced by activated B cells (plasma cells). They have variable regions that bind specifically to antigen epitopes, and constant regions that determine their class and role. Their main function is to neutralize, opsonize, or eliminate pathogens through immune mechanisms.

MARKING SCHEME (3 Marks)
- Definition — 1 mark; structure / function — 1 mark; specificity/role — 1 mark.

In contrast, **antigens** are foreign molecules such as proteins, polysaccharides, or lipids that trigger an immune response when detected by the body. They contain specific sites, called epitopes, recognized by antibodies or T cell receptors. Thus, antibodies act as effector molecules of the adaptive immune system, while antigens serve as the stimulus for immune activation.

(g) Inborn Immunity and Acquired Immunity

Ans. Inborn (innate) immunity and acquired (adaptive) immunity differ in onset, specificity, and memory. **Inborn immunity** is present at birth and acts as the body's first and non-specific defense. It includes physical barriers (skin, mucous membranes), chemical barriers (lysozyme, stomach acid), and cellular defenses such as phagocytes and natural killer cells. It provides immediate but generalized protection, without memory of past infections.

MARKING SCHEME (3 Marks)
- Definition — 1 mark; specificity/memory — 1 mark; components/response — 1 mark.

In contrast, **acquired immunity** develops after exposure to pathogens or through vaccination. It is highly specific to particular antigens and involves T and B lymphocytes. Acquired immunity also provides long-term protection by forming memory cells, which enable faster and stronger responses on re-exposure. Together, these systems ensure both immediate and lasting defense against infections.

(h) Primary and Secondary Immune Responses

Ans. The **primary immune response** occurs during the first exposure to a specific antigen. It is slow, often taking several days to weeks for B and T lymphocytes to become fully activated. Antibody levels are relatively low and short-lived, and symptoms of illness are more noticeable since the immune system requires time to build defense. Memory cells are formed during this phase but do not act immediately.

MARKING SCHEME (3 Marks)
- Occurrence/onset — 1 mark; antibody levels/memory — 1 mark; symptoms — 1 mark.

In contrast, the **secondary immune response** occurs upon subsequent exposure to the same antigen. It is much faster, often within hours, and produces higher and longer-lasting levels of antibodies. Memory B and T cells are reactivated, ensuring efficient pathogen clearance. As a result, symptoms are milder or absent.

III Long Answer Questions

Q.1 Describe the human skin as an impenetrable barrier against invasion by microbes.

Ans. Please see SLO [B-12-1-01]

Q.2 What is the role of macrophages and neutrophils in killing bacteria?

Ans. Please see SLO [B-12-1-05]

Q.3 Explain the protective proteins of complementary system with diagram.

Ans. *Please see SLO* [B-12-1-07]

Q.4 Explain in detail inflammatory response with diagram.

Ans. Please see SLO [B-12-1-08]

Q.5 Explain temperature response against infection.

Ans. Please see SLO [B-12-1-09]

Q.6 What are the ways fever kills microbes?

Ans. Please see SLO [B-12-1-10]

Q.7 Describe the role of monocytes in immune system.

Ans. Please see SLO [B-12-1-1]

Q.8 What is the role of T cells in cell mediated response.

Ans. Please see SLO [B-12-1-15]

Q.9 Describe the role of B cells in humoral antibody mediated immune response.

Ans. Please see SLO [B-12-1-16]

Q.10 Describe the types of acquired immunity.

Ans. Please see SLO [B-12-1-14]

Q.11 Describe the role of T cells and B cells in transplant rejection.

Ans. Please see SLO [B-12-1-17]

Q.12 Explain why a transplant recipient is given immune suppressant drugs and determine what implications this has on his life.

Ans. Please see SLO [B-12-1-17]

Q.13 Evaluate the discovery of monoclonal antibodies and justify how this accomplishment revolutionized many aspects of biological research.

Ans. Please see SLO [B-12-1-18]